

ALMA MATER STUDIORUM – UNIVERSITÀ DI BOLOGNA

DEPARTMENT OF STATISTICAL SCIENCES  
“PAOLO FORTUNATI”

Two-Years Master in Quantitative Finance



ALMA MATER STUDIORUM  
UNIVERSITÀ DI BOLOGNA

# Machine Learning-Based State Modeling for Dynamic Investment Strategies

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Academic Year 2025/2026

## Abstract

In recent years, Machine Learning has become increasingly relevant in quantitative finance as a tool for extracting structure from complex information sets and supporting systematic investment decisions. Within this context, this thesis develops a machine-learning-based framework for dynamic portfolio allocation in which macroeconomic information is organized into latent information states and subsequently translated into portfolio decisions.

The framework considers two alternative feature representations: the original standardized variables (RAW) and a lower-dimensional representation obtained through Principal Component Analysis (PCA). K-Means clustering is used to identify partitions of the information space, while Linear Discriminant Analysis (LDA) is employed in the baseline framework to estimate the probability that new observations belong to the identified information states.

State-specific portfolios are then constructed through Maximum Sharpe Ratio and Mean–Variance optimization. The baseline framework is evaluated under Static and Expanding configurations, while additional experiments consider a Rolling PCA specification and an Annual Adaptive Re-estimation framework. All strategies are assessed through out-of-sample backtesting and compared with conventional market and no-clustering benchmarks using return, risk, risk-adjusted performance and CAPM measures, net of proportional transaction costs.

The empirical results show that several information-state-based strategies achieve competitive out-of-sample risk-adjusted performance while maintaining lower volatility and drawdowns than the main equity benchmarks. The Expanding framework provides the strongest baseline results, while the Rolling PCA specification increases responsiveness and the Adaptive framework offers a more robust decision process by avoiding permanent reliance on a single model configuration.

The main contribution of this thesis is therefore the development and empirical evaluation of a transparent and replicable Machine Learning framework capable of transforming macroeconomic information into systematic and dynamically updated portfolio allocation decisions.

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# Introduction

## Background

Over the past decades, quantitative finance has undergone a profound transformation driven by the rapid growth of data availability, computational power and algorithmic methods. In particular, Machine Learning has evolved from a research topic into a standard component of modern quantitative investment processes, supporting applications ranging from asset pricing and risk management to portfolio construction and algorithmic trading.

The increasing availability of financial, economic and alternative data has enabled the development of increasingly sophisticated models capable of extracting information from complex and high-dimensional datasets. At the same time, the widespread adoption of open-source programming languages and scientific computing libraries has considerably lowered the technological barriers required to develop and evaluate quantitative investment models.

Despite these developments, much of the existing literature and many institutional investment frameworks rely on proprietary datasets, specialized infrastructures or methodologies that are difficult to replicate outside large financial institutions. Consequently, there remains growing interest in understanding whether publicly available data, combined with established Machine Learning techniques, can provide sufficiently informative representations of the underlying information to support systematic investment decisions.

Within this context, this thesis proposes a fully reproducible Machine Learning framework that transforms publicly available information into alternative data representations and evaluates their effectiveness through dynamic investment strategies.

## Research Objective

The primary objective of this thesis is to investigate how different Machine Learning configurations influence the construction of dynamic investment strategies when relying exclusively on publicly available information.

To address this objective, a unified framework is developed to transform heterogeneous economic, financial, geopolitical, energy and uncertainty-related indicators into alternative representations of the available information. Two different approaches are considered throughout the analysis: one based on the original variables (RAW) and another relying on Principal Component Analysis (PCA) for dimensionality reduction.

These alternative representations are subsequently partitioned using the K-Means clus-

tering algorithm, while Linear Discriminant Analysis (LDA) is employed to classify new observations into the identified states. The resulting information is then incorporated into portfolio allocation decisions through alternative optimization criteria.

Rather than focusing on a single model specification, the proposed framework systematically evaluates multiple configurations by varying the data representation, the number of clusters, the portfolio optimization criterion and the model updating methodology. The performance of each configuration is assessed through an out-of-sample backtesting procedure in order to identify the combinations that provide the most effective balance between return and risk.

## Model Selection Criterion

The proposed framework adopts a systematic model comparison approach rather than relying on a single predefined specification. Instead of assuming that one particular configuration is universally optimal, multiple alternative implementations are evaluated to assess how different methodological choices influence investment performance.

Finally, different degrees of model re-estimation are investigated. The baseline framework is evaluated under both Static and Expanding configurations, while two additional extensions are considered: a Rolling PCA framework, in which the macroeconomic representation and clustering structure are recursively re-estimated, and an Annual Adaptive Re-estimation framework, in which the model specification itself is periodically selected using historically available information. The performance of each configuration is evaluated through an out-of-sample backtesting procedure using a comprehensive set of return, risk and risk-adjusted performance measures. This comparative framework makes it possible to identify the combinations of methodological choices that lead to the most effective dynamic investment strategies while ensuring that all evaluations are conducted under a consistent experimental setting.

## Contribution

This thesis contributes to the literature on Machine Learning applications in quantitative finance by proposing a unified and fully reproducible framework for the construction and evaluation of dynamic investment strategies based exclusively on publicly available information.

Rather than introducing a new Machine Learning algorithm, the study provides a systematic comparison of alternative methodological choices within a common experimental setting. Specifically, it investigates the impact of different data representations, clustering granularities, portfolio optimization criteria and model updating procedures on the performance of dynamic investment strategies.

By evaluating all model configurations under the same out-of-sample backtesting framework, the analysis isolates the contribution of each methodological component and provides empirical evidence on how different design choices influence investment performance.

Beyond its empirical findings, the proposed framework is designed to be transparent, modular and easily extendable, providing a foundation for future research on the integration of Machine Learning techniques into systematic portfolio management.

# Chapter 1

## Literature Review

### 1.1 Machine Learning in Quantitative Finance

Over the last two decades, the rapid growth in data availability, computational power and storage capacity has profoundly transformed quantitative finance. While traditional econometric models remain fundamental tools for financial analysis, the increasing complexity, dimensionality and heterogeneity of modern financial and macroeconomic datasets have fostered the widespread adoption of Machine Learning techniques as complementary tools for extracting information and supporting investment decisions [3, 5, 14].

Unlike conventional statistical models, which are typically developed under predefined distributional assumptions, Machine Learning algorithms are primarily data-driven and aim to identify patterns directly from the available observations. This characteristic allows them to capture complex, nonlinear and high-dimensional relationships that may be difficult to model using traditional econometric techniques. Consequently, Machine Learning has become an increasingly important component of quantitative finance, finding applications in asset pricing, return forecasting, volatility modelling, credit risk assessment, algorithmic trading and portfolio management [5, 12, 14].

Machine Learning methodologies are generally divided into two broad categories: supervised and unsupervised learning. Supervised learning relies on labelled observations to estimate the relationship between explanatory variables and a target variable, making it particularly suitable for prediction and classification tasks. Typical financial applications include default prediction, credit scoring, fraud detection and return forecasting. Conversely, unsupervised learning operates in the absence of predefined labels and aims to uncover hidden structures or similarities within the data. Clustering algorithms partition observations into relatively homogeneous groups according to their statistical characteristics, while dimensionality reduction techniques seek compact representations of high-dimensional datasets by preserving their most informative features.

Recent research has increasingly focused on hybrid frameworks that integrate both learning paradigms. In these approaches, unsupervised techniques are first employed to extract latent structures from the data, which are subsequently exploited by supervised models for prediction or probabilistic classification. Such frameworks combine the exploratory capabilities of unsupervised learning with the predictive power of supervised algorithms, providing flexible and data-driven solutions to a wide range of financial prob-

lems [4, 8].

The growing adoption of Machine Learning has therefore expanded the set of quantitative tools available to both researchers and practitioners, enabling the development of increasingly sophisticated models for financial decision-making. The following sections review the literature on market state identification, dimensionality reduction, machine learning techniques for state modelling and portfolio allocation, which together provide the theoretical foundation for the empirical framework developed in this thesis.

## 1.2 Market States and Regime Identification

Financial markets do not evolve under constant economic conditions. Periods of economic expansion, financial stress, elevated inflation or heightened geopolitical uncertainty are often associated with substantially different return distributions, volatility levels and cross-asset correlation structures. Consequently, the assumption that a single statistical model can adequately describe financial markets over long time horizons is frequently unrealistic. This observation has motivated a large body of literature devoted to the identification of market states, with the objective of capturing recurring economic environments characterised by distinct financial dynamics [2, 7].

Early contributions to this literature relied primarily on parametric models. Among these, the Markov-Switching framework introduced by Hamilton [7] represents one of the most influential approaches, allowing model parameters to vary according to an unobservable latent state. Since then, regime-switching methodologies have been widely employed in finance to model changes in asset returns, volatility and business cycle dynamics, demonstrating that the statistical properties of financial markets are not constant over time [2, 6].

The existence of different market states has important implications for portfolio management. If expected returns, volatilities and cross-asset correlations vary across economic environments, portfolio allocations should adapt accordingly rather than relying on a single unconditional distribution of returns. This idea has motivated the development of regime-based asset allocation strategies, in which investment decisions are conditioned on the estimated state of the market. Empirical evidence suggests that dynamically adjusting portfolio allocations across different market environments may improve the risk-return trade-off relative to static allocation strategies [6, 16].

More recently, the increasing availability of high-dimensional financial and macroeconomic datasets has encouraged the adoption of data-driven approaches for market state identification. Rather than imposing a specific probabilistic model on the evolution of regimes, these methodologies infer recurring states directly from the observed data. Clustering algorithms have become one of the most widely adopted techniques in this context, as they partition observations according to their statistical similarity without requiring

predefined labels or strong distributional assumptions.

This perspective is particularly suitable when the objective is not to identify the business cycle itself, but rather to extract information from a broad set of economic indicators that may be relevant for investment decisions. In this context, the resulting states should not necessarily be interpreted as formal economic regimes; instead, they represent recurring information states characterised by similar macroeconomic conditions. The relevance of these information states ultimately depends on their ability to discriminate periods associated with different asset-return dynamics.

The framework proposed in this thesis follows this data-driven approach. Information states are identified from a set of publicly available macroeconomic, financial and uncertainty indicators and subsequently employed to support dynamic portfolio allocation. The following sections review the machine learning techniques adopted to construct these information states and their role within the proposed investment framework.

### 1.3 Dimensionality Reduction for Financial Data

Financial and macroeconomic datasets are often characterised by a large number of variables describing different aspects of the economic environment. In many cases, these variables exhibit strong correlations, resulting in a considerable degree of redundancy within the available information. The presence of highly correlated predictors may complicate subsequent statistical analyses, increase model complexity and reduce the interpretability of the extracted information. Consequently, dimensionality reduction techniques have become an important component of many machine learning frameworks applied to quantitative finance [8, 11].

Among the available techniques, Principal Component Analysis (PCA) represents one of the most widely adopted approaches for transforming a high-dimensional dataset into a smaller set of orthogonal variables while preserving as much variability as possible. Since its introduction by Pearson [17] and its subsequent formalisation by Hotelling [9], PCA has found numerous applications in economics and finance, including yield curve analysis, risk factor extraction, macroeconomic forecasting and asset pricing [11].

The popularity of PCA in financial applications stems from several practical advantages. First, it reduces the dimensionality of the dataset while retaining most of the information contained in the original variables. Second, the resulting components are orthogonal by construction, mitigating potential multicollinearity issues that may affect subsequent modelling stages. Finally, the reduced representation often improves computational efficiency, particularly when machine learning algorithms are applied to datasets containing many correlated features.

Within the context of market state identification, dimensionality reduction may also facilitate the identification of latent structures by concentrating the relevant information

into a limited number of variables. At the same time, reducing the dimensionality of the feature space may improve the stability of clustering algorithms, whose performance can deteriorate in the presence of highly correlated or noisy variables.

For these reasons, dimensionality reduction has become a common preprocessing step in many machine learning applications in finance. Nevertheless, whether it actually improves the quality of the extracted information depends on the characteristics of the available dataset and the objective of the subsequent analysis. Accordingly, this thesis evaluates Principal Component Analysis as an alternative representation of the information set and compares its performance with that obtained using the original standardized variables.

## 1.4 State Identification

The identification of market states has increasingly benefited from the adoption of Machine Learning techniques. While traditional regime-switching models rely on explicit probabilistic assumptions regarding the evolution of latent states, machine learning approaches seek to identify recurring structures directly from the observed data. This data-driven perspective has become particularly attractive in financial applications, where the underlying relationships among economic variables are often complex, nonlinear and difficult to model parametrically.

Among unsupervised learning techniques, clustering algorithms have emerged as one of the most widely adopted approaches for state identification. Their objective is to partition observations into relatively homogeneous groups according to a predefined measure of similarity, without requiring prior information on the underlying state labels. Unlike supervised learning methods, clustering algorithms do not attempt to predict a target variable; instead, they identify recurring patterns that may reveal different market environments or economic conditions. Applications include the identification of business cycle phases, volatility regimes, financial stress episodes and macroeconomic states relevant for asset allocation [1, 10].

Several clustering methodologies have been proposed in the literature, including hierarchical clustering, Gaussian Mixture Models and density-based algorithms. Among these alternatives, K-Means remains one of the most widely used methods because of its computational efficiency, simplicity and interpretability, particularly when the objective is to identify compact and well-separated groups of observations [13, 15]. For these reasons, K-Means has been extensively employed in financial applications involving market segmentation, regime identification and portfolio construction.

Although clustering algorithms provide a partition of the available observations, investment decisions often benefit from a probabilistic representation of the identified states. Financial markets rarely transition abruptly between distinct environments, and periods

of economic uncertainty frequently exhibit characteristics associated with multiple states simultaneously. Several studies have therefore combined unsupervised and supervised learning techniques, using the clusters generated during the unsupervised stage as labels for a subsequent classification model. This approach makes it possible to estimate posterior probabilities of state membership rather than relying exclusively on deterministic assignments, thereby providing a more flexible representation of the prevailing market environment.

The empirical framework proposed in this thesis follows this hybrid perspective. Information states are first extracted through an unsupervised learning algorithm and subsequently incorporated into a supervised classification framework, allowing portfolio allocation decisions to account explicitly for the uncertainty associated with state identification.

## 1.5 Portfolio Allocation under Market States

The identification of market states is not an end in itself, but rather a means of supporting investment decisions. If different states are characterised by distinct return distributions, volatility levels and cross-asset correlation structures, investors may benefit from adapting portfolio allocations to the prevailing market environment instead of relying on a single unconditional investment strategy.

Traditional portfolio allocation models generally assume that expected returns and covariance matrices remain sufficiently stable over the investment horizon. However, a growing body of literature has shown that these quantities may vary substantially across different economic and financial conditions. Consequently, portfolio strategies that explicitly account for changing market states have attracted increasing attention in both academic research and professional asset management [2, 6].

Regime-based portfolio allocation consists of estimating state-dependent investment strategies and adjusting portfolio exposures according to the estimated prevailing state. The underlying intuition is that assets may exhibit different risk-return characteristics across market environments. For example, equities typically perform better during expansionary periods, while government bonds and other defensive assets often provide protection during episodes of elevated uncertainty or financial stress. Conditioning portfolio decisions on the current market state therefore offers the potential to improve risk-adjusted performance relative to static allocation strategies.

Recent developments in regime-based portfolio management have increasingly focused on evaluating alternative portfolio construction frameworks rather than relying on a single investment strategy. Different state representations, portfolio optimization criteria and updating procedures may all influence the resulting asset allocation and, ultimately, the performance of the investment strategy. Consequently, the effectiveness of a regime-based framework should be assessed empirically through a systematic comparison of competing

model specifications rather than inferred solely from the state identification process.

The empirical analysis developed in this thesis follows this perspective. Rather than constructing a single portfolio allocation, the proposed framework generates multiple investment strategies by combining alternative feature representations, state identification models and portfolio optimization approaches. Each configuration produces a distinct sequence of portfolio allocations, whose performance is subsequently evaluated through an out-of-sample backtesting procedure. The resulting strategies are compared with one another and against traditional market benchmarks using a comprehensive set of performance and risk measures, allowing the contribution of each modelling choice to be assessed empirically.

# Chapter 2

## Data

### 2.1 Data Sources

The empirical analysis combines two complementary categories of publicly available data. The first consists of economic, financial and uncertainty indicators used to identify recurring information states, while the second comprises financial market data employed for portfolio construction and performance evaluation.

The use of publicly available data represents a deliberate design choice of the proposed framework. Rather than relying on proprietary databases or institutional data providers, all information employed throughout the analysis can be freely accessed by researchers and practitioners. This ensures that the proposed methodology is fully reproducible and allows its implementation without requiring access to commercial financial databases.

Table 2.1 summarizes the data sources used in the empirical analysis.

Table 2.1: Data Sources

| <b>Data</b>                                | <b>Source</b>                        |
|--------------------------------------------|--------------------------------------|
| Macroeconomic and financial indicators     | Federal Reserve Economic Data (FRED) |
| Geopolitical Risk Index (GPR)              | Caldara and Iacoviello Database      |
| Global Supply Chain Pressure Index (GSCPI) | Federal Reserve Bank of New York     |
| Financial asset prices                     | Yahoo Finance                        |

The following sections describe the selected variables, the investment universe and the preprocessing procedures applied before constructing the machine learning framework.

### 2.2 Macroeconomic and Information Variables

The identification of information states requires variables capable of capturing different dimensions of the economic environment. To this end, the empirical analysis considers a set of publicly available indicators related to energy markets, inflation, financial conditions, credit supply, economic uncertainty, geopolitical risk and global supply-chain disruptions.

The selected variables are reported in Table 2.2.

Table 2.2: Macroeconomic and Information Variables

| Variable     | Description                             | Source                           |
|--------------|-----------------------------------------|----------------------------------|
| WTI          | West Texas Intermediate Crude Oil Price | FRED                             |
| GAS          | Henry Hub Natural Gas Price             | FRED                             |
| ENERGY_CPI   | Energy Consumer Price Index             | FRED                             |
| FIN_STRESS   | St. Louis Financial Stress Index        | FRED                             |
| BANK_LENDING | Senior Loan Officer Survey              | FRED                             |
| WUI          | World Uncertainty Index                 | FRED                             |
| WUPI         | World Pandemic Uncertainty Index        | FRED                             |
| GPR          | Geopolitical Risk Index                 | Caldara & Iacoviello             |
| GSCPI        | Global Supply Chain Pressure Index      | Federal Reserve Bank of New York |

The selected indicators are intended to provide a broad representation of the information available to investors regarding the prevailing economic environment. Energy-related variables (WTI, GAS and ENERGY\_CPI) capture commodity market dynamics and inflationary pressures, while FIN\_STRESS and BANK\_LENDING describe financial conditions and credit availability within the economy.

The information set is further enriched by variables measuring different forms of uncertainty. The World Uncertainty Index (WUI) and the World Pandemic Uncertainty Index (WUPI) capture global uncertainty arising from economic and health-related events, whereas the Geopolitical Risk Index (GPR) reflects the intensity of geopolitical tensions based on newspaper coverage. Finally, the Global Supply Chain Pressure Index (GSCPI) provides a synthetic measure of disruptions affecting global production and transportation networks.

Taken together, these variables describe complementary dimensions of the macro-financial environment and constitute the information set employed throughout the empirical analysis to identify recurring information states.

## 2.3 Investment Universe

The empirical analysis is conducted on a diversified investment universe composed of exchange-traded funds (ETFs). Compared with individual securities, ETFs provide broad market exposure, high liquidity and relatively low transaction costs, making them suitable

instruments for evaluating dynamic asset allocation strategies. Their widespread availability also contributes to the transparency and reproducibility of the proposed framework.

The selected assets are reported in Table 2.3.

Table 2.3: Investment Universe

| <b>Ticker</b> | <b>Asset Class</b>                    |
|---------------|---------------------------------------|
| SPY           | U.S. Equity Market (S&P 500)          |
| QQQ           | U.S. Technology Equities (Nasdaq-100) |
| VGK           | European Equities                     |
| EEM           | Emerging Market Equities              |
| TLT           | Long-Term U.S. Treasury Bonds         |
| GLD           | Gold                                  |
| DBC           | Broad Commodity Basket                |
| BIL           | U.S. Treasury Bills (Risk-Free Proxy) |

The investment universe provides diversified exposure across the principal asset classes considered in strategic asset allocation. Equities are represented through U.S., European and emerging markets, while fixed income is captured by long-term U.S. Treasury bonds. Exposure to alternative asset classes is obtained through gold and a broad commodity basket, allowing the portfolio to incorporate assets that have historically exhibited different behaviour across economic environments. Finally, short-term U.S. Treasury bills are included as a proxy for the risk-free asset and are used both in the portfolio optimization process and in the computation of risk-adjusted performance measures.

The selected assets constitute the investment universe over which all portfolio allocation strategies are evaluated. Their daily returns are subsequently combined with the monthly information states identified by the machine learning framework, allowing the empirical analysis to assess whether different information states translate into different portfolio allocation decisions and, ultimately, different investment performance relative to conventional benchmark strategies.

## 2.4 Data Preprocessing

The variables employed in the empirical analysis are available at different sampling frequencies, including monthly, weekly and quarterly observations. Before constructing the machine learning dataset, all variables are aligned to a common monthly frequency in order to obtain a consistent information set.

Variables originally available at a monthly frequency are retained without modifying their sampling frequency. The St. Louis Financial Stress Index (FIN\_STRESS), which is

published weekly, is converted to monthly observations by computing the average value within each month. Quarterly indicators, namely BANK\_LENDING, WUI and WUPI, are aligned to the monthly frequency using a last observation carried forward (LOCF) procedure, whereby each published observation is maintained until a new value becomes available. Consequently, no interpolation is performed during the frequency alignment process.

Once the sampling frequencies have been harmonized, the variables are transformed where appropriate. WTI, GAS and ENERGY\_CPI are converted into month-over-month percentage changes in order to capture short-term movements in energy prices and inflation. The remaining variables are retained in levels, as they already represent standardized indices or survey-based indicators whose absolute values are economically meaningful.

Finally, all variables are synchronized using a common monthly time index, producing the information set employed throughout the empirical analysis.

## 2.5 Sample Construction

The empirical sample begins in March 2006. This starting date is determined by the joint availability of the selected macroeconomic and information variables and of all ETFs included in the investment universe. Restricting the analysis to the common availability period avoids the use of synthetic or substitute financial series and ensures that all portfolio strategies are evaluated on the same set of investable assets.

The dataset is divided into two non-overlapping periods. The in-sample period extends from May 2007 to December 2021 and is used for the initial estimation of the machine learning models and portfolio allocation parameters. The out-of-sample period begins in January 2022 and extends to the end of april 2026. This period is reserved for the evaluation of the investment strategies.

The separation between the in-sample and out-of-sample periods prevents future observations from entering the initial estimation stage. The precise use of the information available within each period, together with the updating rules adopted under the alternative model configurations, is described in Chapter 3.

# Chapter 3

## Methodology

### 3.1 Overall Framework

The proposed methodology combines unsupervised learning, supervised learning and portfolio optimization into a unified framework for dynamic asset allocation. Starting from the macroeconomic information set described in Chapter 2, the objective is to identify recurring information states and exploit them to construct state-dependent investment strategies.

Following the preprocessing stage, two alternative feature representations are considered. The first consists of the original standardized variables (RAW), while the second is obtained through Principal Component Analysis (PCA). Evaluating both representations allows the impact of dimensionality reduction on the subsequent stages of the framework to be assessed.

The selected feature representation is subsequently employed to identify recurring information states through the K-Means clustering algorithm. The resulting clusters provide a data-driven partition of the macroeconomic information space and constitute the basis for the remaining stages of the methodology.

Although K-Means can assign new observations to the nearest cluster centroid, this assignment is deterministic and does not quantify the uncertainty associated with the classification. For this reason, the cluster labels generated during the unsupervised learning stage are subsequently employed to train a Linear Discriminant Analysis (LDA) classifier. Rather than producing only a single state assignment, the LDA classifier estimates the posterior probability that each observation belongs to every information state, allowing portfolio allocations to evolve smoothly as the estimated macroeconomic environment changes over time.

The identified information states evolve at a monthly frequency, consistently with the frequency of the macroeconomic information set. During the in-sample period, each month is assigned to one of the identified information states. For every state, all daily asset returns observed during the months belonging to that state are collected and used to estimate two state-specific portfolio allocations, corresponding to the Maximum Sharpe Ratio and Mean–Variance optimization criteria. Consequently, each information state is associated with two optimal asset allocations, one for each optimization criterion. These portfolios constitute the investment rules employed throughout the out-of-sample period.

Once the state-specific portfolios have been estimated, newly observed macroeconomic information is used to determine the prevailing information state through the trained LDA classifier. The posterior probabilities estimated by the classifier are then used to obtain the portfolio allocation implemented at the beginning of each month through a probability-weighted combination of the state-specific portfolios. The resulting allocation is held throughout the corresponding month, while portfolio performance is evaluated using the realized daily returns of the selected assets. Consequently, monthly information states determine the portfolio allocation, whereas portfolio returns continue to evolve at a daily frequency.

Depending on the model configuration, different components of the framework may remain fixed or be updated as new information becomes available. Under the Static framework, both the machine learning models and the state-specific portfolio allocations remain unchanged throughout the out-of-sample period. Under the Expanding framework, the feature representation, clustering structure and classification model remain fixed, while the state-specific portfolio parameters are progressively updated.

Two additional adaptive configurations are also considered. The Rolling PCA framework re-estimates the standardization parameters, the PCA transformation and the K-Means clustering model at each monthly rebalancing date using an expanding information set. Consequently, both the feature representation and the corresponding information-state structure are allowed to evolve over time. The Annual Re-estimation framework periodically updates the complete machine learning pipeline, including the feature representation, clustering structure, classification model and portfolio allocation rules.

Finally, the proposed framework is evaluated through a comprehensive out-of-sample backtesting procedure. The empirical implementation considers multiple combinations of feature representations, numbers of information states, portfolio optimization approaches and model updating schemes. Their performance is subsequently compared with conventional market benchmarks in order to evaluate the effectiveness and robustness of the proposed methodology. The implementation details and empirical results are presented in the following chapter.

## 3.2 Simplifying Assumptions

The proposed framework relies on a number of simplifying assumptions intended to ensure a consistent and reproducible empirical analysis. While these assumptions do not capture every aspect of real-world portfolio management, they allow the contribution of the proposed methodology to be evaluated without introducing additional sources of complexity. The main assumptions adopted throughout the analysis are summarized below.

### 3.2.1 Availability of Macroeconomic Information

The framework assumes that all macroeconomic and information variables are available at the beginning of each month, allowing the prevailing information state to be estimated before portfolio rebalancing. In practice, several indicators are released with publication delays and may subsequently be revised. These effects are not explicitly modelled, and the analysis therefore assumes that the complete monthly information set is available when the portfolio allocation is determined.

### 3.2.2 Revised Historical Data

All macroeconomic variables are assumed to be available in their final revised form. Consequently, the analysis does not distinguish between the values initially released to market participants and those subsequently revised by the data providers. Although the use of real-time vintages would provide a more realistic representation of the information available to investors, revised historical data allow the empirical analysis to focus on the effectiveness of the proposed methodology.

### 3.2.3 Transaction Costs

Portfolio turnover is penalized through proportional transaction costs equal to 10 basis points at each portfolio rebalancing. The same transaction cost is applied to every asset and remains constant throughout the sample period. Asset-specific bid-ask spreads, liquidity effects and other time-varying execution costs are not considered.

### 3.2.4 Market Frictions

Apart from the transaction costs described above, financial markets are assumed to be frictionless. The empirical analysis therefore neglects market impact, taxes, liquidity constraints, slippage and execution delays, assuming that all transactions are executed immediately at the observed market prices.

## 3.3 Feature Representation

Following the preprocessing stage described in Chapter 2, the macroeconomic information set is transformed into the feature space employed by the machine learning framework. Since the quality of the input representation may substantially influence the subsequent clustering and classification results, two alternative feature representations are considered throughout the empirical analysis.

The first representation, referred to as *RAW*, consists of the standardized macroeconomic and information variables without any further transformation. This approach

preserves the original economic interpretation of each variable while allowing the clustering algorithm to exploit the complete information content of the dataset.

The second representation is obtained by applying Principal Component Analysis (PCA) to the standardized variables. As discussed in Chapter 1, PCA constructs a new set of orthogonal variables that summarize the common variation of the original dataset while reducing its dimensionality. Rather than representing individual macroeconomic indicators, the principal components capture the dominant sources of variation within the information set.

Both feature representations are analysed independently throughout the empirical analysis. In each case, the resulting feature space serves as the input to the K-Means clustering algorithm, allowing the impact of dimensionality reduction on the identification of information states and the subsequent portfolio allocation process to be evaluated.

To avoid information leakage, all preprocessing transformations are estimated exclusively using the in-sample observations. The parameters of the StandardScaler and, when applicable, the PCA loading matrix are estimated only on the training sample and subsequently applied unchanged to the out-of-sample observations. This procedure ensures that no future information enters the feature construction stage.

### 3.3.1 Standardization

Before constructing the feature space, all variables are standardized in order to eliminate differences in measurement scales and ensure that each variable contributes equally to the subsequent machine learning algorithms. For each variable, the standardized observation is computed as

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}, \quad (3.1)$$

where  $x_{ij}$  denotes the value of variable  $j$  for observation  $i$ , while  $\mu_j$  and  $\sigma_j$  represent the sample mean and standard deviation of variable  $j$ , respectively.

The standardization parameters are estimated exclusively on the training sample and subsequently applied unchanged to the out-of-sample observations, preventing information leakage.

### 3.3.2 Principal Component Analysis

Principal Component Analysis (PCA) transforms the standardized variables into a new set of orthogonal variables, referred to as principal components, that capture the dominant sources of variation within the original dataset.

Given the standardized data matrix  $\mathbf{Z}$ , PCA is based on the sample covariance matrix

$$\Sigma = \frac{1}{n-1} \mathbf{Z}^\top \mathbf{Z}, \quad (3.2)$$

whose eigenvectors and eigenvalues satisfy

$$\Sigma \mathbf{v}_k = \lambda_k \mathbf{v}_k, \quad (3.3)$$

where  $\lambda_k$  denotes the variance explained by the  $k$ -th principal component and  $\mathbf{v}_k$  is the corresponding eigenvector.

The principal components are then obtained as

$$\mathbf{PC} = \mathbf{ZV}, \quad (3.4)$$

where  $\mathbf{V}$  is the matrix containing the selected eigenvectors.

### 3.3.3 Selection of the Number of Principal Components

The number of retained principal components is determined according to the cumulative explained variance criterion. The contribution of the  $k$ -th principal component is measured through its explained variance ratio,

$$EV_k = \frac{\lambda_k}{\sum_{j=1}^p \lambda_j}, \quad (3.5)$$

where  $\lambda_k$  denotes the eigenvalue associated with the  $k$ -th principal component and  $p$  is the total number of original variables. The cumulative explained variance is then computed as

$$CEV_m = \sum_{k=1}^m EV_k. \quad (3.6)$$

The selection of the number of principal components involves a trade-off between dimensionality reduction and information preservation. Retaining a limited number of components allows the original feature space to be represented more parsimoniously, while increasing the number of retained components preserves a larger proportion of the variability contained in the original dataset.

The explained variance and cumulative explained variance therefore provide useful diagnostic measures for evaluating how much of the original information is preserved by alternative lower-dimensional representations. In practical applications, the final number of retained components can be selected by considering both the proportion of variance explained and the desired degree of dimensionality reduction.

Once the number of components has been determined, the selected principal components constitute the reduced-dimensional feature representation employed in the subsequent stages of the analysis.

## 3.4 State Identification

The standardized observations are partitioned into homogeneous information states through the K-Means clustering algorithm. Given a predefined number of clusters, the algorithm iteratively assigns each observation to the nearest centroid and updates the centroid locations until convergence.

The clustering problem is formulated as

$$\min_C \sum_{k=1}^K \sum_{\mathbf{x}_i \in C_k} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2, \quad (3.7)$$

where  $C_k$  denotes the set of observations belonging to cluster  $k$ ,  $\boldsymbol{\mu}_k$  is the corresponding centroid and  $\mathbf{x}_i$  is the feature vector associated with observation  $i$ .

The clustering model is estimated exclusively on the training sample in order to prevent information leakage. The resulting clusters define a partition of the macroeconomic information space and provide the state labels employed in the subsequent classification stage.

### 3.4.1 Selection of the Number of Information States

The number of information states is not known a priori. Consequently, four alternative specifications are considered throughout the empirical analysis, corresponding to two, three, four and five clusters.

Rather than selecting a single configuration before the empirical analysis, all specifications are retained and independently combined with the alternative feature representations, portfolio optimization approaches and model updating schemes. Their relative performance is subsequently evaluated through the out-of-sample backtesting results presented in Chapter 4.

## 3.5 State Classification

The cluster labels obtained during the previous stage are subsequently employed to estimate a Linear Discriminant Analysis (LDA) classifier. The objective is not to redefine the information states, but to learn the relationship between the selected feature representation and the clusters identified by the K-Means algorithm.

Once estimated on the training sample, the classifier is applied to newly observed macroeconomic information during the out-of-sample period. Although new observations could also be assigned to the nearest cluster centroid, such an approach provides only a deterministic classification. Instead, the LDA classifier estimates the posterior probability that each observation belongs to every information state, allowing the uncertainty

associated with the classification to be explicitly incorporated into the portfolio allocation process.

The classifier is estimated exclusively using the training sample and remains unchanged throughout the baseline analysis. Consequently, the estimated posterior probabilities constitute the link between the state identification stage and the subsequent portfolio construction procedure.

### 3.5.1 Probabilistic State Representation

For each observation, the LDA classifier assumes that the feature vectors belonging to each information state follow a multivariate normal distribution with a common covariance matrix,

$$f_k(\mathbf{x}) = \frac{1}{(2\pi)^{p/2} |\boldsymbol{\Sigma}|^{1/2}} \exp \left[ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}_k) \right], \quad (3.8)$$

where  $\boldsymbol{\mu}_k$  denotes the mean vector of information state  $k$ ,  $\boldsymbol{\Sigma}$  is the covariance matrix shared across all information states and  $p$  is the number of features.

The posterior probability that an observation belongs to information state  $k$  is then obtained through Bayes' theorem,

$$P(G = k \mid \mathbf{x}) = \frac{\pi_k f_k(\mathbf{x})}{\sum_{j=1}^K \pi_j f_j(\mathbf{x})}, \quad (3.9)$$

where  $\pi_k$  denotes the prior probability of information state  $k$ .

Unlike a hard classification approach, the posterior probabilities provide a probabilistic representation of the prevailing macroeconomic environment. Rather than assigning each observation to a single state, they allow multiple information states to contribute simultaneously to the portfolio allocation according to their estimated probabilities.

## 3.6 Portfolio Construction

The final stage of the proposed methodology consists of associating each identified information state with an optimal portfolio allocation.

For every information state identified during the training period, all daily asset returns observed within the corresponding months are collected. These observations are subsequently used to estimate the expected return vector and covariance matrix required by the portfolio optimization procedure. The optimization is performed independently for each information state and for each optimization criterion considered in the empirical analysis. Consequently, each information state is associated with a state-specific optimal portfolio.

The state-specific portfolios represent the investment rules learned from the historical relationship between macroeconomic information states and asset returns. Their construction is entirely based on the training sample and provides the foundation for the out-of-sample allocation procedure described in the following sections.

### 3.6.1 Portfolio Optimization

For each information state, the corresponding daily asset returns are used to estimate a state-specific expected return vector,  $\boldsymbol{\mu}_k$ , and covariance matrix,  $\boldsymbol{\Sigma}_k$ . These quantities constitute the inputs to the portfolio optimization procedures and are estimated independently for every information state. Consequently, each information state is associated with its own optimal portfolio allocation.

The Mean–Variance optimization problem for information state  $k$  is formulated as

$$\max_{\mathbf{w}_k} \left( \mathbf{w}_k^\top \boldsymbol{\mu}_k - \frac{\lambda}{2} \mathbf{w}_k^\top \boldsymbol{\Sigma}_k \mathbf{w}_k \right), \quad (3.10)$$

where  $\mathbf{w}_k$  denotes the portfolio weights associated with information state  $k$ ,  $\boldsymbol{\mu}_k$  is the corresponding expected return vector,  $\boldsymbol{\Sigma}_k$  is the associated covariance matrix and  $\lambda$  is the investor’s risk-aversion parameter. Throughout the empirical analysis, the risk-aversion parameter is fixed at  $\lambda = 10$ . This value is kept constant across all model configurations to ensure a consistent comparison between the alternative portfolio construction approaches.

As an alternative, the Maximum Sharpe Ratio portfolio for information state  $k$  is obtained by solving

$$\max_{\mathbf{w}_k} \frac{\mathbf{w}_k^\top (\boldsymbol{\mu}_k - r_{f,k})}{\sqrt{\mathbf{w}_k^\top \boldsymbol{\Sigma}_k \mathbf{w}_k}}, \quad (3.11)$$

where  $r_{f,k}$  denotes the state-specific risk-free rate, estimated as the average return of the BIL over the observations assigned to information state  $k$ .

Both optimization problems are solved independently for every information state under the following constraints,

$$\sum_{i=1}^N w_{i,k} = 1, \quad (3.12)$$

$$0 \leq w_{i,k} \leq 1, \quad i = 1, \dots, N, \quad (3.13)$$

thereby imposing fully invested, long-only portfolios without leverage or short selling.

### 3.6.2 Probability-Weighted Portfolio Allocation

The optimization procedure described above produces one optimal portfolio for each information state. During the out-of-sample period, however, the prevailing macroeconomic environment is not represented by a deterministic state assignment but by the posterior probabilities estimated by the LDA classifier.

Rather than selecting the portfolio associated with the most likely information state, the proposed framework combines all state-specific portfolios according to their estimated posterior probabilities. The portfolio implemented at each rebalancing date is therefore obtained as

$$\mathbf{w} = \sum_{k=1}^K p_k \mathbf{w}_k, \quad (3.14)$$

where  $p_k$  denotes the posterior probability of belonging to information state  $k$  and  $\mathbf{w}_k$  is the corresponding state-specific optimal portfolio.

This probability-weighted allocation provides a smooth transition between different macroeconomic environments, avoiding abrupt changes in portfolio composition while explicitly incorporating the uncertainty associated with the state classification. Consequently, all identified information states contribute to the investment decision in proportion to their estimated likelihood.

### 3.6.3 Portfolio Turnover and Transaction Costs

Portfolio rebalancing generates trading activity whenever the portfolio weights change over time. To account for this effect, the empirical analysis incorporates proportional transaction costs at each portfolio rebalancing date.

Let  $\mathbf{w}_t$  denote the vector of portfolio weights implemented at time  $t$ . Portfolio turnover is measured as the total absolute variation in portfolio weights between two consecutive rebalancing dates,

$$\text{Turnover}_t = \sum_{i=1}^N |w_{i,t} - w_{i,t-1}|, \quad (3.15)$$

where  $w_{i,t}$  denotes the portfolio weight allocated to asset  $i$  after the rebalancing at time  $t$ , and  $N$  is the number of assets.

Assuming proportional transaction costs equal to  $c$ , the total trading cost incurred at each rebalancing date is computed as

$$TC_t = c \times \text{Turnover}_t, \quad (3.16)$$

where throughout the empirical analysis the transaction cost parameter is fixed at

$$c = 10 \text{ basis points} = 0.001.$$

The resulting net portfolio return is therefore computed as

$$r_t^{\text{net}} = r_t^{\text{gross}} - TC_t. \quad (3.17)$$

Including transaction costs allows the empirical analysis to evaluate whether the proposed dynamic allocation framework remains effective after accounting for a more realistic trading frictions. Moreover, the probability-weighted portfolio allocation described in the previous subsection contributes to smoother changes in portfolio weights, thereby limiting portfolio turnover relative to strategies based on abrupt switches between information states.

### 3.6.4 Investment Timing

Before describing the alternative model configurations, it is important to clarify the timing convention adopted in the empirical analysis and the distinction between the in-sample estimation stage and the out-of-sample investment procedure.

During the in-sample period, each monthly macroeconomic observation is associated with the information state identified for the same month. The daily asset returns observed within that month are then assigned to the corresponding information state and used to estimate the state-specific expected returns, covariance matrices and portfolio allocations. This contemporaneous association is used exclusively for learning the historical relationship between macroeconomic information states and asset returns.

During the out-of-sample period, a stricter investment timing convention is adopted. The macroeconomic information observed for month  $t$  is used to determine the portfolio allocation implemented during month  $t + 1$ . Specifically, once the information associated with month  $t$  becomes available, it is processed by the relevant machine learning model and translated into the portfolio weights to be held over the following investment month.

Therefore, the portfolio return observed during month  $t + 1$  is generated using an allocation based exclusively on information available up to month  $t$ . Formally,

$$\mathbf{w}_{t+1} = g(\mathcal{I}_t), \quad (3.18)$$

where  $\mathcal{I}_t$  denotes the information set available at the end of month  $t$ , and  $g(\cdot)$  represents the portfolio construction procedure associated with the selected model configuration.

This timing convention prevents future macroeconomic information from entering the out-of-sample investment decision and therefore preserves the predictive nature of the backtesting exercise.

### 3.6.5 Alternative Model Configurations

The proposed framework is evaluated under four alternative model configurations. All configurations adopt the general portfolio construction procedure described in the previous sections, but differ in the components that are updated as new information becomes available throughout the out-of-sample period.

#### Static Framework

The Static framework represents the baseline specification of the proposed methodology. The feature representation, K-Means clustering model, LDA classifier and state-specific portfolio allocations are estimated exclusively using the training sample and remain unchanged throughout the entire out-of-sample period.

At each monthly rebalancing date, the newly available macroeconomic information is processed by the fixed LDA classifier to estimate the posterior probabilities associated with the information states. These probabilities are subsequently combined with the state-specific portfolios estimated during the training phase to determine the portfolio allocation for the following investment period.

Consequently, no model parameter is updated after the initial estimation. Portfolio allocations evolve exclusively as a result of changes in the estimated posterior probabilities, while the machine learning model and the state-specific optimal portfolios remain fixed throughout the out-of-sample analysis.

#### Expanding Framework

The Expanding framework preserves the feature representation, K-Means clustering model and LDA classifier estimated on the training sample, while allowing the portfolio optimization stage to incorporate newly available information.

The feature representation itself is therefore kept unchanged throughout the entire out-of-sample period. New macroeconomic observations are standardized using the mean and standard deviation estimated on the original training sample and, when the PCA representation is adopted, projected onto the same principal-component loading matrix estimated in-sample. Consequently, the feature space remains fixed over time, while only the state-specific portfolio optimization is updated as additional return observations become available.

At each monthly rebalancing date, the newly available macroeconomic observations are processed by the fixed LDA classifier, which estimates the posterior probabilities associated with the existing information states. Each new observation is then assigned to the information state with the highest posterior probability, while the state definitions themselves remain unchanged.

The daily asset returns corresponding to the newly classified month are subsequently added to the historical return sample of the assigned information state. The expected return vectors, covariance matrices and state-specific risk-free rates are then re-estimated using the expanded dataset, resulting in updated state-specific portfolio allocations that incorporate the newly available information.

This configuration allows the portfolio optimization stage to adapt progressively as new observations become available, while preserving a fixed representation of the macroeconomic information states throughout the entire out-of-sample period.

### **Rolling PCA Framework**

The Rolling PCA framework investigates whether continuously updating the macroeconomic feature representation improves the portfolio construction process. Unlike the baseline specification, the StandardScaler, the PCA transformation and the K-Means clustering model are re-estimated at every monthly rebalancing date using an expanding window containing all macroeconomic observations available up to the end of the previous month.

Consequently, both the principal-component representation and the corresponding information states are allowed to evolve over time as new macroeconomic information becomes available. At each monthly re-estimation, the most recent macroeconomic observation is assigned to one of the updated information states, which determines the portfolio implemented during the following investment month.

The portfolio optimization stage is then performed using only the historical daily asset returns associated with the selected information state and available before the current rebalancing date. State-specific expected returns, covariance matrices and the risk-free rate are therefore recomputed every month before solving the Maximum Sharpe and Mean–Variance optimization problems.

Unlike the baseline framework, which combines the state-specific portfolios using the posterior probabilities estimated by the LDA classifier, the Rolling PCA specification implements a single portfolio corresponding to the information state identified by the monthly clustering procedure. This experiment therefore evaluates whether continuously updating the macroeconomic representation and the identification of information states leads to a more adaptive investment strategy.

### **Annual Adaptive Re-estimation Framework**

The Annual Adaptive Re-estimation framework periodically updates the complete machine learning and portfolio construction pipeline using an expanding historical sample.

At the beginning of each calendar year, all macroeconomic and financial information available up to the end of the previous year is incorporated into the training sample.

The StandardScaler, the PCA transformation (when applicable), the K-Means clustering model and the LDA classifier are then re-estimated using the expanded dataset. Subsequently, all candidate portfolio specifications are reconstructed under the updated information states.

Rather than adopting a fixed portfolio specification throughout the entire out-of-sample period, the framework evaluates the out-of-sample performance achieved by every candidate model during the previous calendar year. The portfolio specification exhibiting the highest Sharpe ratio is then selected and implemented during the subsequent year. Consequently, the model specification itself becomes adaptive over time, allowing both the regime identification procedure and the portfolio construction methodology to evolve according to their historical out-of-sample performance.

Unlike the Static and Expanding frameworks, this approach updates not only the state-specific portfolio weights but also the underlying model specification, thereby providing an adaptive mechanism capable of responding to structural changes in the macroeconomic environment.

## 3.7 Performance Metrics

The performance of the proposed portfolio allocation framework is evaluated using a comprehensive set of performance indicators covering return, risk, downside risk and market-adjusted performance. These measures allow the different portfolio configurations to be compared not only in terms of profitability, but also with respect to their overall risk profile and their exposure to systematic market risk.

### 3.7.1 Return and Risk Measures

#### Annual Return

Portfolio performance is evaluated using the annualized return computed from the cumulative portfolio return over the entire evaluation period. Let  $V_0$  and  $V_T$  denote the initial and final portfolio values, respectively. The annualized return is computed as

$$R_{\text{ann}} = 252 \bar{r}, \quad (3.19)$$

where  $\bar{r}$  denotes the average daily portfolio return over the evaluation period, so that  $R_{\text{ann}}$  represents the annualized average portfolio return.

#### Annual Volatility

Portfolio risk is measured through the annualized standard deviation of daily portfolio returns,

$$\sigma_{\text{ann}} = \sqrt{252} \text{Std}(r_t), \quad (3.20)$$

where  $r_t$  denotes the daily portfolio return computed from consecutive closing prices.

### Sharpe Ratio

Risk-adjusted performance is evaluated through the Sharpe Ratio,

$$SR = \frac{R_{\text{ann}} - R_f}{\sigma_{\text{ann}}}, \quad (3.21)$$

where  $R_f$  denotes the annualized risk-free rate.

### Maximum Drawdown

The maximum drawdown measures the largest cumulative loss experienced by the portfolio from a historical peak,

$$MDD = \min_t \left( \frac{V_t - \max_{s \leq t} V_s}{\max_{s \leq t} V_s} \right), \quad (3.22)$$

where  $V_t$  denotes the cumulative portfolio value.

### Calmar Ratio

The Calmar Ratio relates annualized return to the maximum drawdown,

$$Calmar = \frac{R_{\text{ann}}}{|MDD|}. \quad (3.23)$$

## 3.7.2 Downside Risk Measures

Tail risk is assessed through the Value-at-Risk (VaR) and the Expected Shortfall (ES), both computed from the empirical distribution of daily portfolio returns.

### Value-at-Risk

For a confidence level  $\alpha$ , the Value-at-Risk is defined as

$$VaR_\alpha = Q_{1-\alpha}(r), \quad (3.24)$$

where  $Q_\alpha(r)$  denotes the empirical  $\alpha$ -quantile of daily portfolio returns.

## Expected Shortfall

The Expected Shortfall measures the expected portfolio loss conditional on returns exceeding the corresponding Value-at-Risk,

$$ES_\alpha = E[r \mid r \leq VaR_\alpha]. \quad (3.25)$$

### 3.7.3 CAPM Performance Measures

To evaluate the market-adjusted performance of the proposed strategies, the Capital Asset Pricing Model (CAPM) is estimated for each portfolio using the QQQ ETF as the market benchmark. The regression model is specified as

$$r_t - r_{f,t} = \alpha + \beta(r_{QQQ,t} - r_{f,t}) + \varepsilon_t, \quad (3.26)$$

where  $r_t$  denotes the daily portfolio return,  $r_{QQQ,t}$  is the daily return of the QQQ ETF and  $r_{f,t}$  is the daily risk-free rate. The estimated regression provides four complementary performance indicators: alpha, beta, the alpha t-statistic and the coefficient of determination ( $R^2$ ).

#### Annual Alpha

The alpha coefficient corresponds to the intercept of the CAPM regression and measures the abnormal portfolio return after controlling for systematic market risk. Formally, the intercept is estimated as

$$\hat{\alpha} = \overline{(r_t - r_{f,t})} - \hat{\beta} \overline{(r_{QQQ,t} - r_{f,t})}, \quad (3.27)$$

where  $\overline{(r_t - r_{f,t})}$  denotes the average portfolio excess return and  $\overline{(r_{QQQ,t} - r_{f,t})}$  is the average excess return of the QQQ benchmark.

Since the empirical analysis reports annual performance measures, the estimated daily alpha is subsequently annualized

$$\alpha_{\text{ann}} = (1 + \hat{\alpha}_{\text{daily}})^{252} - 1, \quad (3.28)$$

where  $\hat{\alpha}_{\text{daily}}$  is the daily CAPM intercept estimated from the regression.

A positive alpha indicates that the portfolio generates returns exceeding those implied by its systematic exposure to the benchmark.

#### Beta

The beta coefficient measures the sensitivity of portfolio excess returns to changes in the excess return of the benchmark portfolio.

$$\hat{\beta} = \frac{\text{Cov}(r_t - r_{f,t}, r_{QQQ,t} - r_{f,t})}{\text{Var}(r_{QQQ,t} - r_{f,t})}. \quad (3.29)$$

A beta equal to one indicates that the portfolio exhibits the same systematic risk as the benchmark, whereas values below one imply lower market exposure.

### Alpha t-statistic

The statistical significance of the estimated alpha is assessed through its t-statistic,

$$t_\alpha = \frac{\hat{\alpha}}{\text{SE}(\hat{\alpha})}, \quad (3.30)$$

where  $\text{SE}(\hat{\alpha})$  denotes the standard error of the estimated alpha. Larger absolute values provide stronger statistical evidence that the portfolio generates abnormal performance.

### Coefficient of Determination

The coefficient of determination is computed as

$$R^2 = 1 - \frac{\sum_{t=1}^T \hat{\varepsilon}_t^2}{\sum_{t=1}^T (y_t - \bar{y})^2}, \quad (3.31)$$

where  $\hat{\varepsilon}_t$  denotes the residual of the CAPM regression,  $y_t = r_t - r_{f,t}$  is the portfolio excess return and  $\bar{y}$  is its sample mean.

The coefficient of determination measures the proportion of the variance of portfolio excess returns explained by the linear relationship with the QQQ excess return.

# Chapter 4

## Empirical Results

### 4.1 Model Comparison

This section presents the empirical results obtained from the baseline version of the proposed framework. As described in Chapter 3, the baseline analysis evaluates the Static and Expanding model configurations under two alternative feature representations (RAW and PCA), four alternative numbers of information states and two portfolio optimization criteria.

Tables 4.1 and 4.2 summarize the corresponding out-of-sample performance. All reported results are net of proportional transaction costs equal to 10 basis points applied at each portfolio rebalancing.

In addition to the conventional market benchmarks represented by the SPY ETF, the QQQ ETF and the Equal Weight portfolio, two additional benchmark strategies are considered. These correspond to Mean–Variance and Maximum Sharpe optimizations estimated exclusively using the in-sample financial returns, without conditioning the portfolio construction process on the identified information states. The resulting portfolio weights are subsequently held fixed throughout the out-of-sample period. Referred to as *BM\_MeanVariance\_NoClusters* and *BM\_Max\_Sharpe\_NoClusters*, these benchmarks isolate the contribution of the information-state identification process by comparing the proposed framework with conventional portfolio optimization based on the same historical information set but without the clustering stage.

Table 4.1 reports the principal performance measures, including annual return, annualized volatility, Sharpe ratio, Value-at-Risk, Expected Shortfall, Maximum Drawdown and Calmar ratio. Table 4.2 complements the analysis by presenting the CAPM statistics computed using the QQQ ETF as the market benchmark, namely annual alpha, beta, alpha t-statistic and the coefficient of determination ( $R^2$ ).

Table 4.1: Out-of-sample performance — return, risk and ratios (net of 10 bps transaction costs)

| Features  | Clusters | Strategy                   | Annual Return | Annual Vol. | Risk-Free | Sharpe  | VaR 95% | VaR 99% | ES 95%  | ES 99%  | Max DD  | Calmar |
|-----------|----------|----------------------------|---------------|-------------|-----------|---------|---------|---------|---------|---------|---------|--------|
| PCA       | 3        | Expanding_Max_Sharpe       | 0.1382        | 0.1403      | 0.0383    | 0.7122  | -0.0147 | -0.0220 | -0.0195 | -0.0288 | -0.2308 | 0.5988 |
| PCA       | 4        | Expanding_Max_Sharpe       | 0.1273        | 0.1405      | 0.0383    | 0.6337  | -0.0140 | -0.0224 | -0.0196 | -0.0289 | -0.2315 | 0.5500 |
| RAW       | 3        | Expanding_Mean_Variance    | 0.1166        | 0.1250      | 0.0383    | 0.6266  | -0.0123 | -0.0204 | -0.0174 | -0.0263 | -0.2058 | 0.5667 |
| RAW       | 3        | Static_Mean_Variance       | 0.1153        | 0.1251      | 0.0383    | 0.6153  | -0.0125 | -0.0204 | -0.0174 | -0.0263 | -0.2058 | 0.5599 |
| Benchmark | -        | Equal Weight               | 0.1116        | 0.1257      | 0.0383    | 0.5834  | -0.0125 | -0.0204 | -0.0174 | -0.0263 | -0.2085 | 0.5352 |
| PCA       | 3        | Static_Mean_Variance       | 0.1114        | 0.1257      | 0.0383    | 0.5815  | -0.0125 | -0.0204 | -0.0174 | -0.0263 | -0.2085 | 0.5341 |
| PCA       | 3        | Expanding_Mean_Variance    | 0.1114        | 0.1257      | 0.0383    | 0.5815  | -0.0125 | -0.0204 | -0.0174 | -0.0263 | -0.2085 | 0.5341 |
| RAW       | 3        | Expanding_Max_Sharpe       | 0.1195        | 0.1400      | 0.0383    | 0.5800  | -0.0145 | -0.0224 | -0.0194 | -0.0291 | -0.2522 | 0.4739 |
| PCA       | 4        | Expanding_Mean_Variance    | 0.1106        | 0.1250      | 0.0383    | 0.5783  | -0.0121 | -0.0204 | -0.0173 | -0.0262 | -0.2082 | 0.5311 |
| PCA       | 4        | Static_Mean_Variance       | 0.1105        | 0.1250      | 0.0383    | 0.5776  | -0.0121 | -0.0204 | -0.0173 | -0.0262 | -0.2082 | 0.5307 |
| PCA       | 2        | Expanding_Max_Sharpe       | 0.1116        | 0.1350      | 0.0383    | 0.5427  | -0.0140 | -0.0223 | -0.0186 | -0.0267 | -0.2390 | 0.4669 |
| PCA       | 2        | Expanding_Mean_Variance    | 0.0974        | 0.1139      | 0.0383    | 0.5187  | -0.0113 | -0.0182 | -0.0155 | -0.0226 | -0.2033 | 0.4789 |
| RAW       | 2        | Expanding_Mean_Variance    | 0.0948        | 0.1139      | 0.0383    | 0.4961  | -0.0113 | -0.0176 | -0.0155 | -0.0228 | -0.2076 | 0.4566 |
| Benchmark | -        | QQQ                        | 0.1535        | 0.2331      | 0.0383    | 0.4944  | -0.0237 | -0.0393 | -0.0331 | -0.0478 | -0.3483 | 0.4408 |
| Benchmark | -        | SPY                        | 0.1254        | 0.1774      | 0.0383    | 0.4913  | -0.0170 | -0.0302 | -0.0254 | -0.0404 | -0.2450 | 0.5121 |
| RAW       | 2        | Expanding_Max_Sharpe       | 0.1035        | 0.1357      | 0.0383    | 0.4806  | -0.0139 | -0.0211 | -0.0186 | -0.0275 | -0.2563 | 0.4038 |
| PCA       | 2        | Static_Max_Sharpe          | 0.0924        | 0.1137      | 0.0383    | 0.4760  | -0.0115 | -0.0180 | -0.0155 | -0.0224 | -0.2069 | 0.4467 |
| RAW       | 2        | Static_Mean_Variance       | 0.0905        | 0.1136      | 0.0383    | 0.4599  | -0.0115 | -0.0178 | -0.0154 | -0.0225 | -0.2094 | 0.4325 |
| RAW       | 5        | Expanding_Mean_Variance    | 0.0908        | 0.1210      | 0.0383    | 0.4336  | -0.0121 | -0.0185 | -0.0165 | -0.0245 | -0.2625 | 0.3457 |
| Benchmark | -        | BM_MeanVariance_NoClusters | 0.0871        | 0.1133      | 0.0383    | 0.4309  | -0.0114 | -0.0177 | -0.0154 | -0.0224 | -0.2114 | 0.4120 |
| PCA       | 5        | Expanding_Mean_Variance    | 0.0859        | 0.1214      | 0.0383    | 0.3920  | -0.0121 | -0.0180 | -0.0165 | -0.0242 | -0.2705 | 0.3175 |
| PCA       | 3        | Static_Max_Sharpe          | 0.0905        | 0.1344      | 0.0383    | 0.3887  | -0.0138 | -0.0209 | -0.0184 | -0.0256 | -0.2391 | 0.3786 |
| RAW       | 4        | Expanding_Mean_Variance    | 0.0808        | 0.1236      | 0.0383    | 0.3439  | -0.0126 | -0.0190 | -0.0171 | -0.0248 | -0.2989 | 0.2704 |
| PCA       | 4        | Static_Max_Sharpe          | 0.0844        | 0.1370      | 0.0383    | 0.3363  | -0.0138 | -0.0215 | -0.0187 | -0.0271 | -0.2399 | 0.3517 |
| PCA       | 5        | Expanding_Max_Sharpe       | 0.0843        | 0.1451      | 0.0383    | 0.3171  | -0.0147 | -0.0224 | -0.0204 | -0.0300 | -0.3060 | 0.2755 |
| RAW       | 5        | Expanding_Max_Sharpe       | 0.0789        | 0.1487      | 0.0383    | 0.2732  | -0.0153 | -0.0233 | -0.0210 | -0.0319 | -0.3184 | 0.2479 |
| RAW       | 3        | Static_Max_Sharpe          | 0.0739        | 0.1346      | 0.0383    | 0.2644  | -0.0138 | -0.0205 | -0.0182 | -0.0259 | -0.2657 | 0.2781 |
| PCA       | 2        | Static_Max_Sharpe          | 0.0667        | 0.1308      | 0.0383    | 0.2171  | -0.0138 | -0.0204 | -0.0179 | -0.0242 | -0.2524 | 0.2643 |
| RAW       | 5        | Static_Mean_Variance       | 0.0625        | 0.1183      | 0.0383    | 0.2045  | -0.0126 | -0.0184 | -0.0160 | -0.0223 | -0.2628 | 0.2378 |
| PCA       | 5        | Static_Mean_Variance       | 0.0612        | 0.1188      | 0.0383    | 0.1930  | -0.0126 | -0.0182 | -0.0160 | -0.0222 | -0.2646 | 0.2313 |
| RAW       | 2        | Static_Max_Sharpe          | 0.0615        | 0.1312      | 0.0383    | 0.1769  | -0.0138 | -0.0204 | -0.0178 | -0.0245 | -0.2690 | 0.2286 |
| Benchmark | -        | BM_Max_Sharpe_NoClusters   | 0.0545        | 0.1333      | 0.0383    | 0.1215  | -0.0140 | -0.0204 | -0.0182 | -0.0250 | -0.2990 | 0.1822 |
| RAW       | 4        | Expanding_Max_Sharpe       | 0.0522        | 0.1427      | 0.0383    | 0.0975  | -0.0149 | -0.0217 | -0.0197 | -0.0273 | -0.3160 | 0.1652 |
| PCA       | 5        | Static_Max_Sharpe          | 0.0270        | 0.1306      | 0.0383    | -0.0865 | -0.0138 | -0.0194 | -0.0178 | -0.0242 | -0.2767 | 0.0976 |
| RAW       | 4        | Static_Max_Sharpe          | 0.0206        | 0.1313      | 0.0383    | -0.1347 | -0.0140 | -0.0191 | -0.0181 | -0.0243 | -0.2950 | 0.0698 |
| RAW       | 5        | Static_Max_Sharpe          | 0.0136        | 0.1299      | 0.0383    | -0.1904 | -0.0140 | -0.0194 | -0.0180 | -0.0243 | -0.2898 | 0.0468 |
| RAW       | 4        | Static_Mean_Variance       | 0.0038        | 0.1307      | 0.0383    | -0.2638 | -0.0138 | -0.0196 | -0.0180 | -0.0242 | -0.3177 | 0.0120 |

Table 4.2: Out-of-sample performance — alpha, beta and CAPM statistics (net of 10 bps transaction costs)

| Features  | Clusters | Strategy                   | Alpha Ann. | Beta vs QQQ | Alpha t-stat | CAPM $R^2$ |
|-----------|----------|----------------------------|------------|-------------|--------------|------------|
| PCA       | 3        | Expanding_Max_Sharpe       | 0.0461     | 0.4753      | 1.0872       | 0.6239     |
| PCA       | 4        | Expanding_Max_Sharpe       | 0.0327     | 0.4937      | 0.8264       | 0.6705     |
| RAW       | 3        | Expanding_Mean_Variance    | 0.0288     | 0.4331      | 0.7986       | 0.6516     |
| RAW       | 3        | Static_Mean_Variance       | 0.0272     | 0.4351      | 0.7589       | 0.6572     |
| Benchmark | -        | Equal Weight               | 0.0226     | 0.4422      | 0.6444       | 0.6725     |
| PCA       | 3        | Static_Mean_Variance       | 0.0224     | 0.4421      | 0.6376       | 0.6723     |
| PCA       | 3        | Expanding_Mean_Variance    | 0.0224     | 0.4421      | 0.6376       | 0.6723     |
| RAW       | 3        | Expanding_Max_Sharpe       | 0.0266     | 0.4765      | 0.6395       | 0.6292     |
| PCA       | 4        | Expanding_Mean_Variance    | 0.0222     | 0.4369      | 0.6273       | 0.6638     |
| PCA       | 4        | Static_Mean_Variance       | 0.0221     | 0.4369      | 0.6248       | 0.6638     |
| PCA       | 2        | Expanding_Max_Sharpe       | 0.0237     | 0.4323      | 0.5413       | 0.5570     |
| PCA       | 2        | Expanding_Mean_Variance    | 0.0220     | 0.3242      | 0.5285       | 0.4401     |
| RAW       | 2        | Expanding_Mean_Variance    | 0.0187     | 0.3298      | 0.4562       | 0.4557     |
| Benchmark | -        | QQQ                        | 0.0000     | 1.0000      | -0.2086      | 1.0000     |
| Benchmark | -        | SPY                        | 0.0032     | 0.7285      | 0.1289       | 0.9158     |
| RAW       | 2        | Expanding_Max_Sharpe       | 0.0145     | 0.4406      | 0.3377       | 0.5730     |
| PCA       | 2        | Static_Max_Sharpe          | 0.0157     | 0.3345      | 0.3906       | 0.4698     |
| RAW       | 2        | Static_Mean_Variance       | 0.0137     | 0.3352      | 0.3424       | 0.4731     |
| RAW       | 5        | Expanding_Mean_Variance    | 0.0135     | 0.3391      | 0.3032       | 0.4269     |
| Benchmark | -        | BM_MeanVariance_NoClusters | 0.0103     | 0.3348      | 0.2580       | 0.4750     |
| PCA       | 5        | Expanding_Mean_Variance    | 0.0090     | 0.3352      | 0.1998       | 0.4145     |
| PCA       | 3        | Static_Max_Sharpe          | -0.0017    | 0.4678      | -0.0438      | 0.6578     |
| RAW       | 4        | Expanding_Mean_Variance    | 0.0008     | 0.3618      | 0.0189       | 0.4656     |
| PCA       | 4        | Static_Max_Sharpe          | -0.0103    | 0.4895      | -0.2830      | 0.6939     |
| PCA       | 5        | Expanding_Max_Sharpe       | 0.0082     | 0.3285      | 0.1372       | 0.2786     |
| RAW       | 5        | Expanding_Max_Sharpe       | 0.0022     | 0.3334      | 0.0361       | 0.2734     |
| RAW       | 3        | Static_Max_Sharpe          | -0.0183    | 0.4694      | -0.4891      | 0.6607     |
| PCA       | 2        | Static_Max_Sharpe          | -0.0207    | 0.4275      | -0.5105      | 0.5804     |
| RAW       | 5        | Static_Mean_Variance       | -0.0137    | 0.3298      | -0.3179      | 0.4220     |
| PCA       | 5        | Static_Mean_Variance       | -0.0159    | 0.3382      | -0.3749      | 0.4409     |
| RAW       | 2        | Static_Max_Sharpe          | -0.0258    | 0.4278      | -0.6344      | 0.5777     |
| Benchmark | -        | BM_Max_Sharpe_NoClusters   | -0.0348    | 0.4480      | -0.8884      | 0.6145     |
| RAW       | 4        | Expanding_Max_Sharpe       | -0.0277    | 0.3640      | -0.5069      | 0.3538     |
| PCA       | 5        | Static_Max_Sharpe          | -0.0499    | 0.3459      | -1.0325      | 0.3812     |
| RAW       | 4        | Static_Max_Sharpe          | -0.0574    | 0.3591      | -1.2110      | 0.4066     |
| RAW       | 5        | Static_Max_Sharpe          | -0.0627    | 0.3471      | -1.3204      | 0.3879     |
| RAW       | 4        | Static_Mean_Variance       | -0.0679    | 0.3105      | -1.3386      | 0.3067     |

### 4.1.1 Discussion of the Baseline Results

The empirical evidence suggests that the proposed framework is capable of generating several portfolio strategies exhibiting an attractive risk–return profile. It is important to emphasize that the objective of the proposed methodology is not to systematically outperform the market in terms of absolute returns. Rather, the purpose is to algorithmically construct portfolio allocations that achieve returns comparable to those of traditional market benchmarks while reducing the overall level of portfolio risk.

From this perspective, the results are encouraging. Although the QQQ benchmark records the highest annual return (15.35%), it is also characterized by the highest annualized volatility (23.31%) and the largest maximum drawdown (34.83%). Similarly, the SPY benchmark achieves an annual return of 12.54% with an annualized volatility of 17.74%. By contrast, many of the algorithmically generated strategies exhibit substantially lower levels of volatility and drawdown while maintaining competitive annual returns.

Among the evaluated specifications, the highest out-of-sample Sharpe ratio is obtained by the PCA representation with three information states under the Expanding framework and Maximum Sharpe optimization. This configuration achieves an annual return of 13.82%, an annualized volatility of 14.03% and a Sharpe ratio of 0.712, representing a more favourable risk–return trade-off than both conventional market benchmarks. Nevertheless, this result should not be interpreted as evidence that this specification will necessarily remain the best-performing model in future periods. Since the methodology is entirely data-driven, the objective is not to identify a single superior strategy ex post, but to evaluate whether the proposed algorithmic framework is capable of systematically generating robust portfolio allocations under different model configurations.

A further result concerns the comparison with the benchmark portfolios constructed without the information-state identification stage. Both *BM\_MeanVariance\_NoClusters* and *BM\_Max\_Sharpe\_NoClusters* are generally outperformed by the majority of the Expanding specifications. This finding suggests that conditioning the portfolio optimization on macroeconomic information states provides additional information that is not captured by conventional optimization methods applied to the full sample.

Finally, the comparison between the Static and Expanding frameworks indicates that updating the portfolio optimization stage as new observations become available generally improves the out-of-sample results. Across both RAW and PCA representations, the best-performing Expanding specifications consistently rank above their corresponding Static counterparts, suggesting that progressively updating the state-specific expected returns and covariance matrices enhances the adaptability of the portfolio allocation while preserving a fixed representation of the information states. These results provide the baseline against which the more dynamic updating schemes introduced in the following sections are evaluated.

## 4.2 Portfolio Construction Process

This section illustrates the main outputs generated by the proposed framework and explains how the identified information states are translated into portfolio allocations. The objective is to provide a financial interpretation of the intermediate results and to illustrate how the estimated information states ultimately determine the portfolio allocation implemented by the investment strategy.

For illustrative purposes, the analysis focuses on the PCA and RAW specifications with three information states. Since the empirical results presented in the previous section indicate that three clusters provide the best overall performance, this configuration is used throughout the following figures to illustrate how the identified information states are translated into portfolio allocations under both the Static and Expanding frameworks.

The discussion proceeds in three steps. First, the principal-component representation of the macroeconomic information set is examined. Second, the resulting information states identified by the clustering algorithm are analysed. Finally, the portfolio allocations generated under the two rebalancing schemes are compared.

### 4.2.1 Principal Component Analysis

The first step of the proposed framework consists of reducing the dimensionality of the standardized macroeconomic information set through Principal Component Analysis (PCA). The objective is to summarize the information contained in the original variables into a smaller set of orthogonal components while preserving most of the total variability. This transformation reduces redundancy among the explanatory variables and provides a more compact representation of the macroeconomic environment to be used in the subsequent clustering procedure.

The PCA model is estimated using only the in-sample observations in order to avoid any look-ahead bias. Once the principal-component loading matrix has been obtained, both the training and the out-of-sample observations are projected onto the same factor space.

Figure 4.1 reports the cumulative explained variance associated with the principal components. The first six components explain more than 90% of the total variability of the standardized macroeconomic dataset, for this reason are retained throughout the empirical analysis in order to preserve a larger fraction of the information contained in the original variables while maintaining a relatively low-dimensional representation of the macroeconomic environment.

The retained principal components constitute the input variables used by the K-Means clustering algorithm in the PCA specification. Their macro-economic meaning is discussed through the cluster centroids presented in the following subsection.

The retained principal components constitute the feature space used by the K-Means

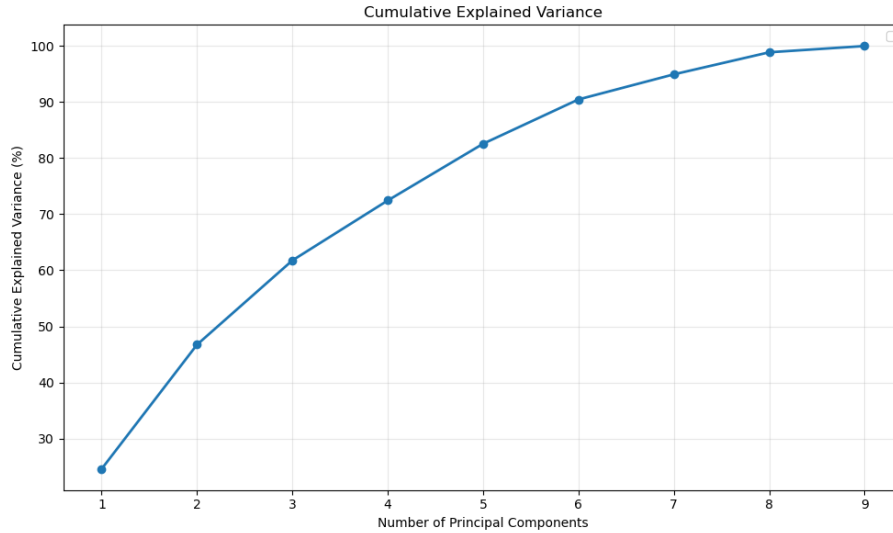


Figure 4.1: Cumulative explained variance explained by the principal components.

clustering algorithm in the PCA specification. The following subsection examines how these latent factors contribute to the identification of the information states by analysing the cluster centroids.

## 4.2.2 Cluster Composition

After constructing the feature representation, the observations are partitioned into information states using the K-Means clustering algorithm. Figure 4.2 reports the standardized centroids obtained under the PCA specification with three information states, while Figure 4.3 presents the corresponding centroids obtained using the RAW macroeconomic variables.

Each row of the heatmap represents the centroid of one information state, whereas each column corresponds to one input variable used by the clustering algorithm. Since all variables are standardized prior to clustering, positive values indicate that the corresponding variable assumes a level above its historical average within that information state, while negative values indicate below-average realizations. Consequently, the heatmaps summarize the characteristic macroeconomic profile associated with each information state rather than the observations themselves.

The PCA specification represents the macroeconomic environment through the first six principal components. Because each principal component is a linear combination of the original variables, the resulting centroids describe the latent factors extracted from the macroeconomic dataset rather than directly observable economic variables. Conversely, the RAW specification preserves the original standardized macroeconomic indicators, allowing the information states to be interpreted directly in terms of the underlying economic variables.

Although the numerical values of the centroids differ across the two specifications, both

approaches aim to identify periods characterized by similar macroeconomic conditions. These information states constitute the basis for the subsequent portfolio construction process, as each state is associated with a different set of expected asset returns and covariance matrices estimated from the corresponding historical observations.

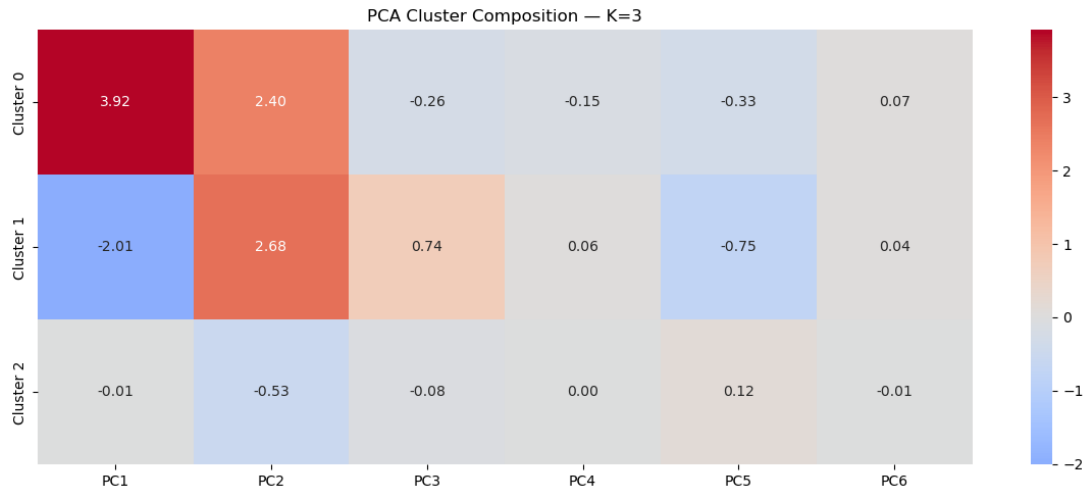


Figure 4.2: Cluster centroids obtained under the PCA specification ( $K = 3$ ).

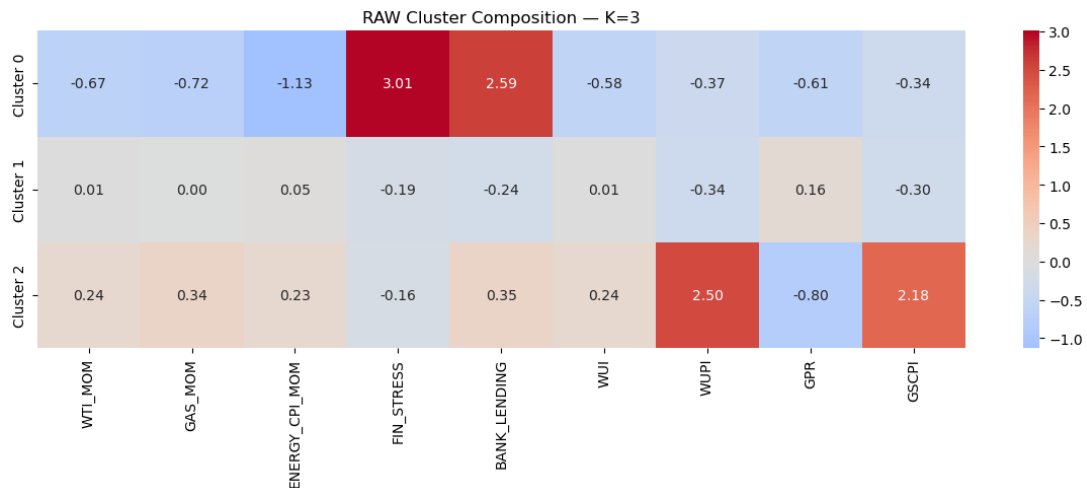


Figure 4.3: Cluster centroids obtained under the RAW specification ( $K = 3$ ).

### 4.2.3 Evolution of Information States

While the cluster centroids describe the characteristics of the information states, it is also useful to examine how the corresponding state assignments evolve over time. Figures 4.4 and 4.5 report the cumulative wealth of the QQQ ETF together with the monthly information states identified under the PCA and RAW specifications, respectively.

The background colour indicates the information state assigned by the K-Means clustering model to each monthly macroeconomic observation. The QQQ series is included

only as a financial reference and does not enter the clustering procedure. Consequently, the figures do not imply that the information states are identified from equity-market performance; rather, they provide a visual representation of how the macroeconomic states identified by the model coincide with different phases of financial-market evolution.

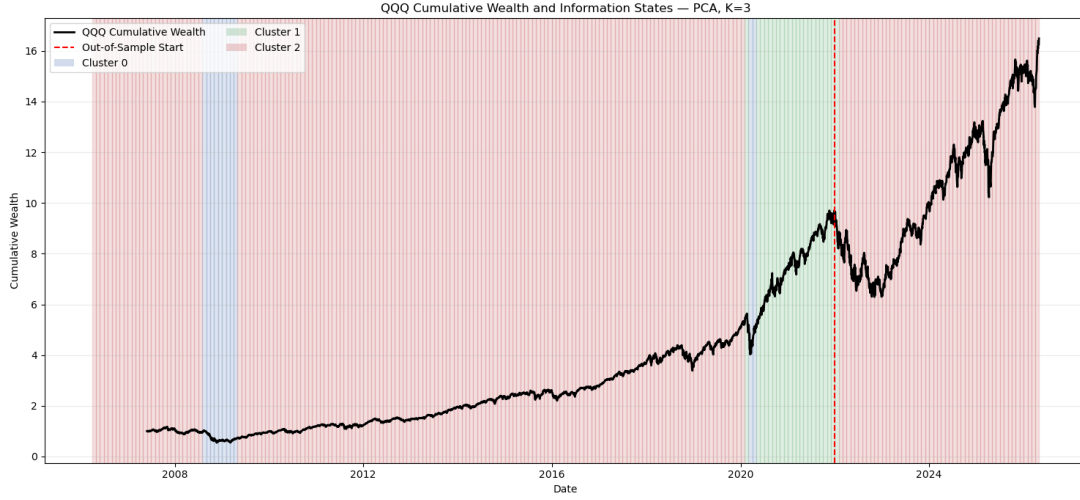


Figure 4.4: QQQ cumulative wealth and monthly information states under the PCA specification ( $K = 3$ ).

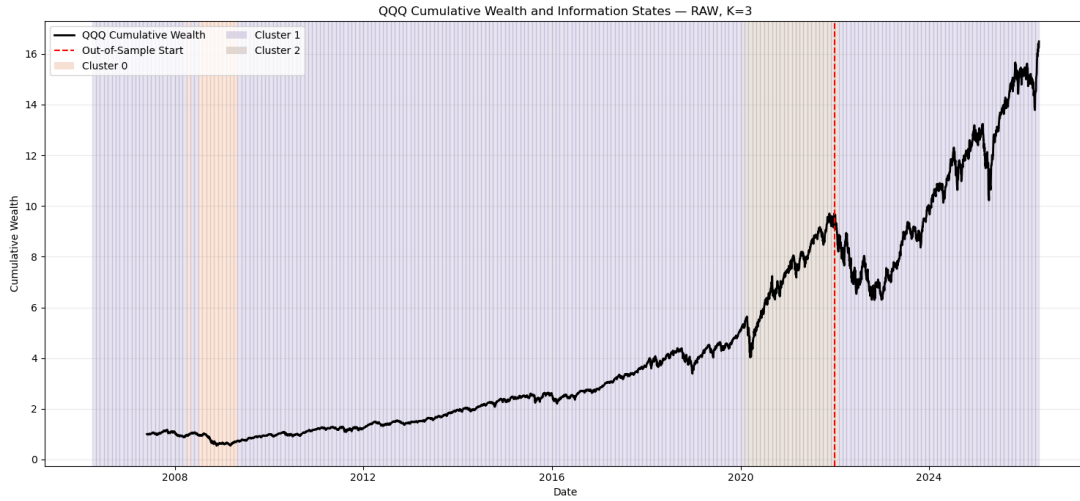


Figure 4.5: QQQ cumulative wealth and monthly information states under the RAW specification ( $K = 3$ ).

#### 4.2.4 Linear Discriminant Analysis

Once the information states have been identified through the K-Means clustering algorithm, Linear Discriminant Analysis (LDA) is employed to estimate the posterior probability that each macroeconomic observation belongs to every information state. Rather than assigning each observation deterministically to a single cluster, the LDA classifier provides a probabilistic assessment of the prevailing financial environment.

The LDA classifier is estimated exclusively using the in-sample observations together with the corresponding K-Means labels. During the out-of-sample period, the estimated classifier remains fixed. At each monthly rebalancing date, the newly available macroeconomic observation is projected onto the feature space estimated during the training period and classified using the previously estimated LDA model. Consequently, the posterior probabilities used for portfolio construction are always computed using only the information available at the time of the investment decision.

Figures 4.6 and 4.7 report the posterior probabilities generated by the LDA classifier under the PCA and RAW specifications, respectively. The upper panels display the in-sample probabilities used to estimate the classifier, whereas the lower panels show the sequential out-of-sample probabilities obtained as new macroeconomic observations become available.

The estimated probabilities are generally concentrated on a single information state, indicating that the classifier is able to distinguish relatively well between the regimes identified by the clustering algorithm. Nevertheless, some observations exhibit more balanced posterior probabilities, reflecting periods in which the prevailing financial environment shares characteristics with multiple information states. Rather than representing abrupt changes in regime, these observations capture the uncertainty associated with the classification process.

Within the proposed framework, these posterior probabilities constitute the link between the macroeconomic information set and the portfolio allocation stage. Under the baseline methodology, the monthly portfolio weights are obtained as the probability-weighted combination of the optimal portfolios associated with each information state.

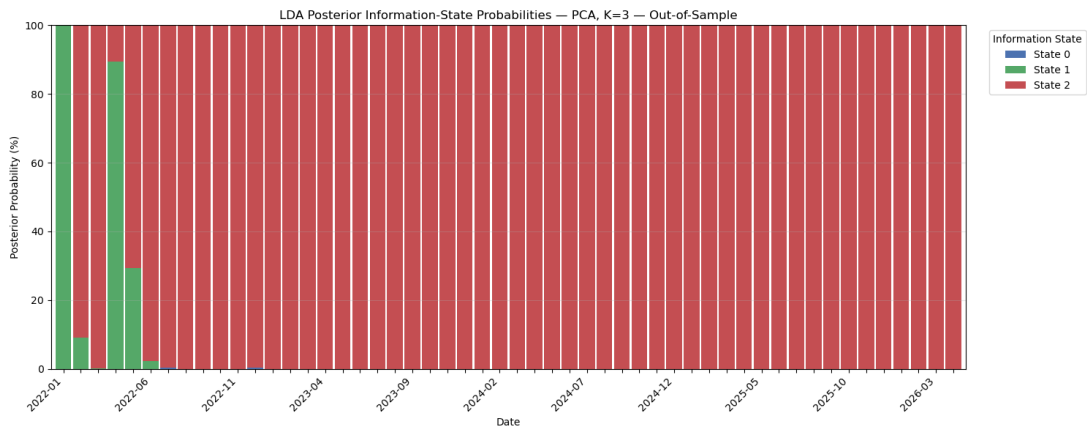


Figure 4.6: Out of sample probabilities under the PCA specification ( $K=3$ )

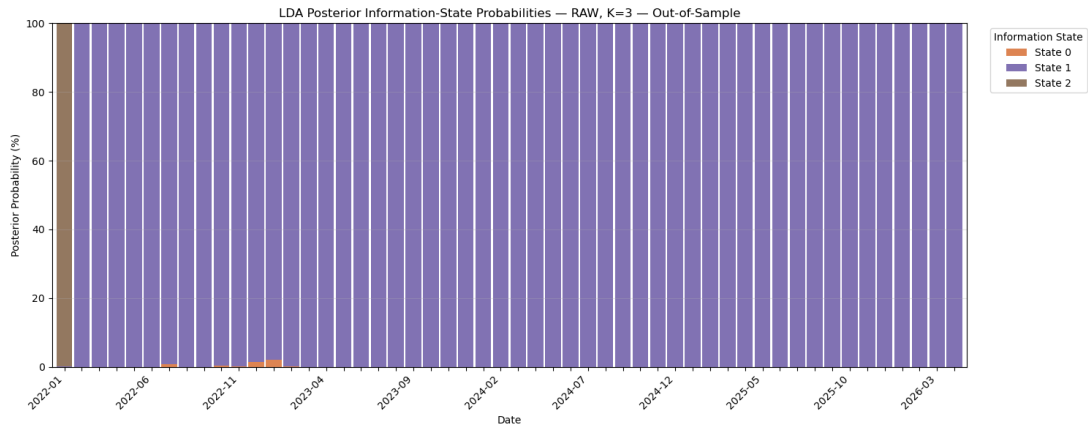


Figure 4.7: Out of sample probabilities under the PCA specification ( $K = 3$ )

The out-of-sample posterior probabilities are generally concentrated on a single information state, indicating that the LDA classifier produces highly confident classifications throughout most of the evaluation period. Only a limited number of observations exhibit more balanced probabilities across multiple states, reflecting periods of greater classification uncertainty. These posterior probabilities constitute the inputs used by the portfolio construction procedure described in the following subsection.

#### 4.2.5 Portfolio Allocation

The final step of the proposed framework consists of translating the estimated information-state probabilities into portfolio allocations. For each information state, an optimal portfolio is estimated using the historical asset returns associated with that state according to either the Maximum Sharpe Ratio or the Mean–Variance optimization criterion.

Under the baseline framework, the monthly portfolio weights are obtained as the probability-weighted average of the state-specific optimal portfolios. Consequently, the evolution of the portfolio allocation reflects both the estimated macroeconomic environment and the characteristics of the optimization criterion adopted.

Under the Static specification, the state-specific portfolios are estimated once using the in-sample observations and remain unchanged throughout the out-of-sample period. Portfolio weights therefore evolve exclusively because the posterior probabilities assigned by the LDA classifier change over time.

Under the Expanding specification, the information states remain fixed, while the expected returns and covariance matrices associated with each state are recursively updated using all financial observations available up to the current rebalancing date. As a result, portfolio weights evolve both because of changes in the estimated information-state probabilities and because of the continuous updating of the optimization inputs.

For the sake of brevity, only the PCA specification with three information states is presented in the following figures. The corresponding results obtained under the RAW

specification are omitted, as they illustrate the same portfolio construction mechanism.

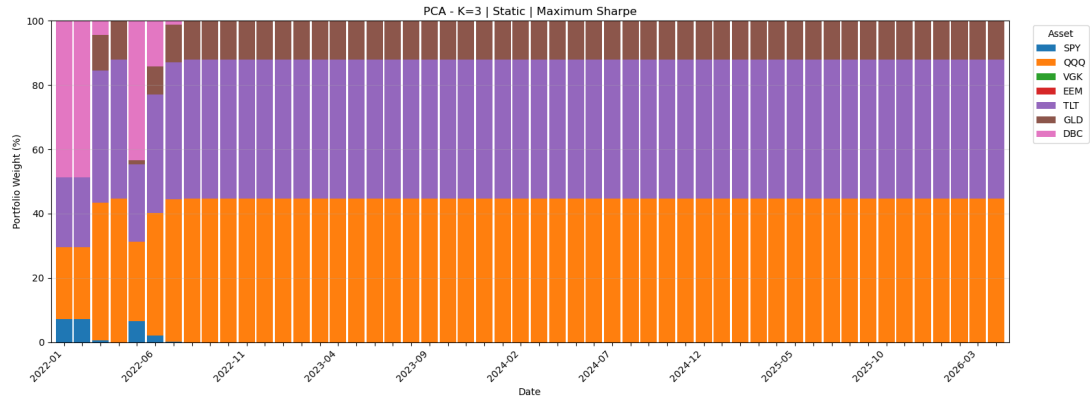


Figure 4.8: Out-of-sample portfolio weights obtained with the static rebalancing approach and the Maximum Sharpe Ratio optimization (PCA specification,  $K = 3$ ).

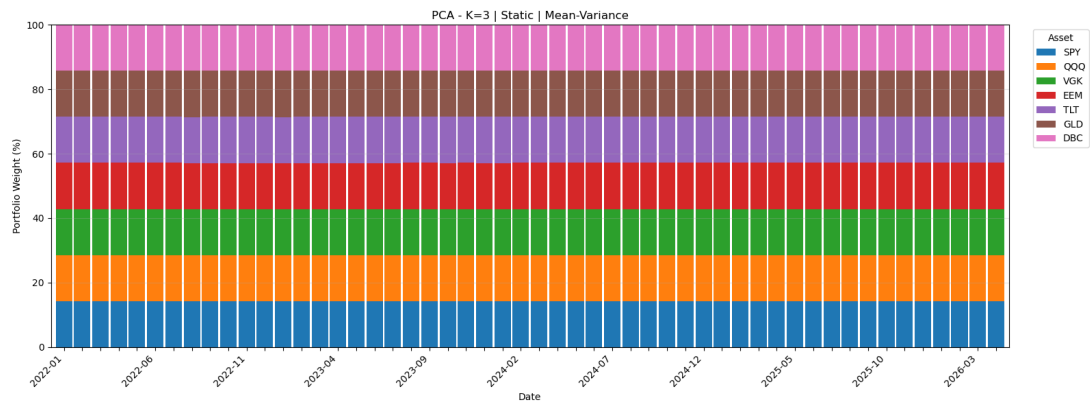


Figure 4.9: Out-of-sample portfolio weights obtained with the static rebalancing approach and the Mean-Variance optimization (PCA specification,  $K = 3$ ).

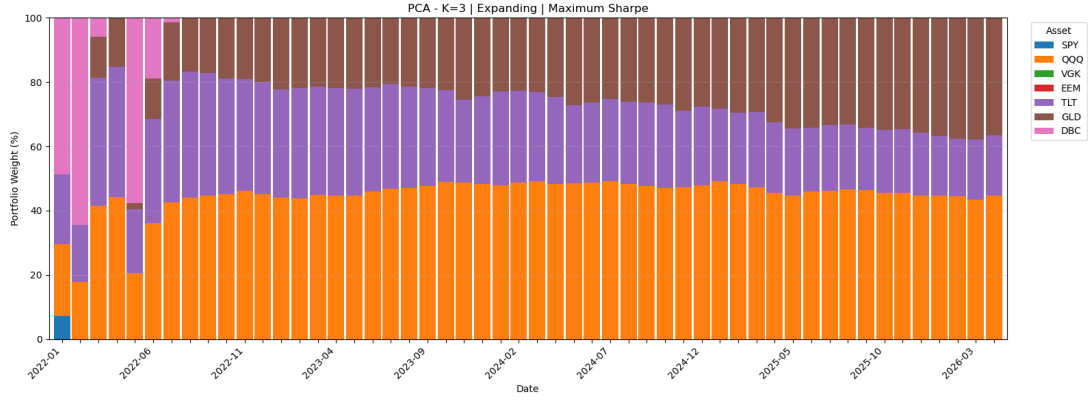


Figure 4.10: Out-of-sample portfolio weights obtained with the expanding-window rebalancing approach and the Maximum Sharpe Ratio optimization (PCA specification,  $K = 3$ ).

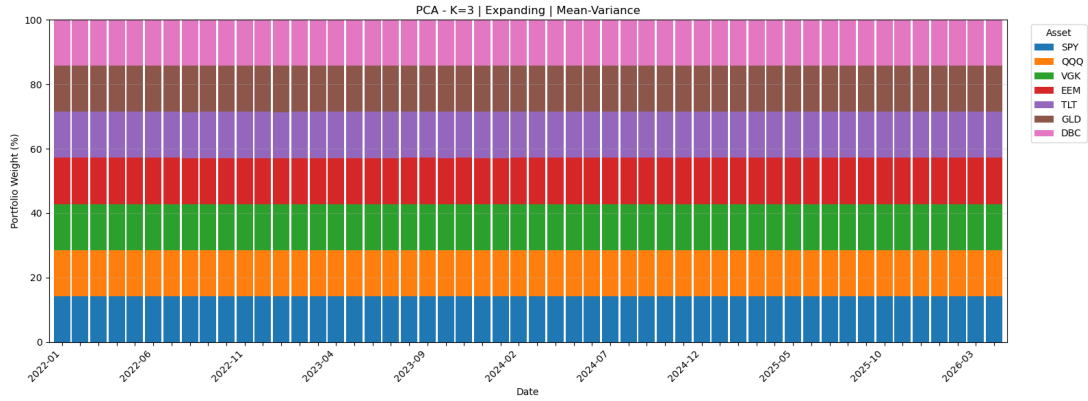


Figure 4.11: Out-of-sample portfolio weights obtained with the expanding-window rebalancing approach and the Mean-Variance optimization (PCA specification,  $K = 3$ ).

## 4.2.6 Cumulative Performance Comparison

Figures 4.12 and 4.13 report the cumulative wealth generated by all regime-based strategies over the out-of-sample period, together with the benchmark portfolios. The first figure refers to the expanding specifications, while the second reports the corresponding static implementations.

A visual inspection suggests that the expanding strategies generally achieve a more favorable evolution of cumulative wealth than their static counterparts. By updating the regime-specific portfolio allocations as new observations become available, the expanding framework appears to adapt more effectively to changing market conditions, particularly during periods characterized by increased market volatility.

Conversely, the static strategies rely on portfolio weights estimated exclusively from the initial training sample and therefore cannot incorporate new information throughout the investment horizon. As a consequence, their cumulative performance tends to be less responsive to structural changes in the underlying market environment.

The cumulative wealth profiles provide a graphical counterpart to the quantitative evidence reported in Section 4.1, highlighting the different out-of-sample dynamics of the expanding and static specifications.

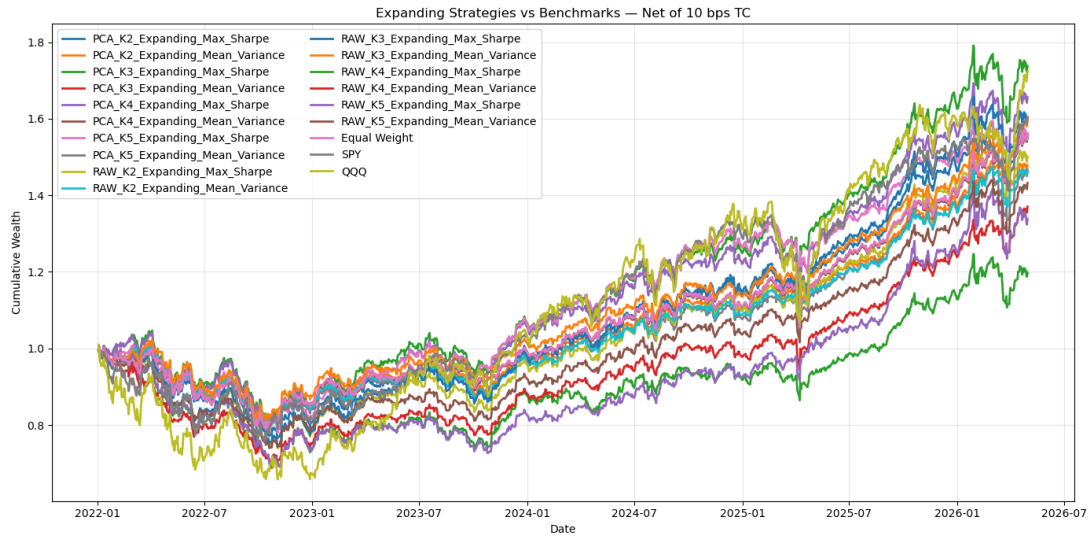


Figure 4.12: Out-of-sample cumulative wealth of the expanding regime-based strategies and benchmark portfolios.

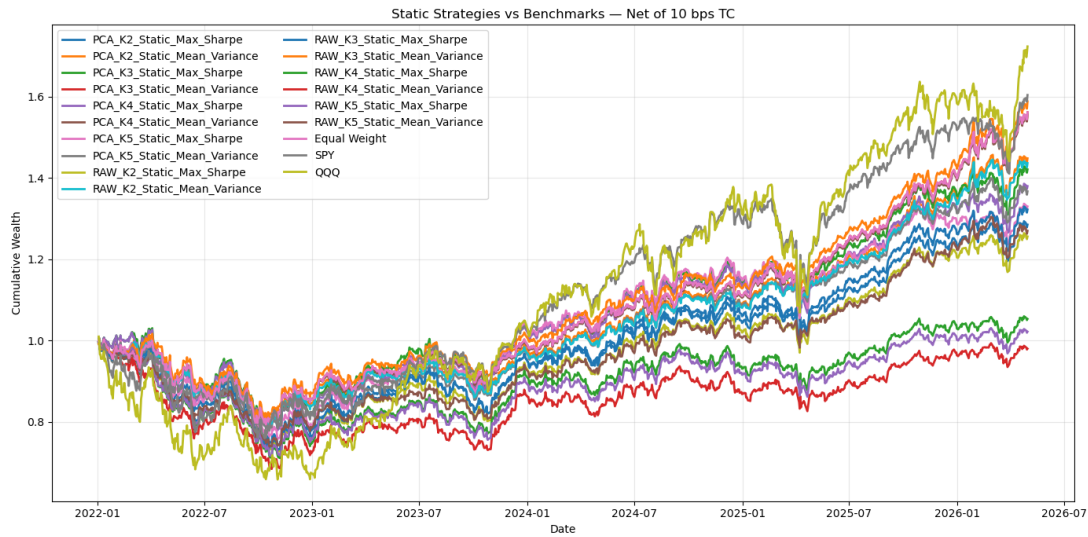


Figure 4.13: Out-of-sample cumulative wealth of the static regime-based strategies and benchmark portfolios.

## 4.2.7 CAPM analysis

The CAPM statistics provide additional insights into the source of the observed portfolio performance. Most of the best-performing Expanding specifications exhibit positive annualized alpha estimates, suggesting that the proposed framework is capable of generating returns exceeding those implied by their systematic exposure to the QQQ benchmark. In contrast, several Static specifications, particularly those based on the Maximum Sharpe optimization criterion, display negative alpha estimates.

This pattern is consistent with the previous finding that periodically updating the state-specific portfolio allocations improves the ability of the strategy to adapt to evolving market conditions. Nevertheless, none of the reported alpha estimates is statistically significant at conventional confidence levels, as the corresponding absolute t-statistics remain well below the critical value of approximately two.

The coefficient of determination further indicates that the CAPM explains a substantial proportion of the variability in the portfolio excess returns. For the best-performing baseline specifications, the reported  $R^2$  values are generally between 0.60 and 0.67, implying that approximately two-thirds of the variation in portfolio returns can be attributed to their systematic exposure to the QQQ benchmark. The remaining variation reflects portfolio-specific components not captured by the single-factor CAPM, suggesting that the proposed framework generates return dynamics that are only partially explained by broad market movements.

Consequently, although the Expanding framework generally produces more favourable abnormal return estimates than the Static specification, the empirical evidence does not support the conclusion that the proposed methodology systematically generates statistically significant abnormal performance after controlling for market risk.

### 4.3 Rolling PCA Framework

The first extension of the baseline methodology considers a dynamically updated principal-component representation. As described in Chapter 3, under the Rolling PCA framework the StandardScaler, the PCA transformation and the K-Means clustering model are re-estimated at each monthly rebalancing date using an expanding information set containing only macroeconomic observations available before the current investment month.

This specification therefore differs from the baseline Expanding framework in an important dimension. While the latter updates the state-specific portfolio parameters while preserving the original feature representation and information-state structure, the Rolling PCA framework also allows the macroeconomic feature space and the corresponding clustering structure to evolve over time.

At each rebalancing date, all historically available macroeconomic observations are re-standardized and projected onto the newly estimated principal-component space. K-Means is subsequently re-estimated within this updated representation, and the most recent available macroeconomic observation determines the information state relevant for the following investment month. Historical financial returns associated with that state are then used to estimate the expected return vector and covariance matrix required by the portfolio optimizer.

Unlike the probability-weighted allocation adopted in the baseline framework, this specification implements the optimal portfolio associated with the information state identified at the current re-estimation date. The procedure is repeated independently for each number of clusters considered in the analysis and for both the Maximum Sharpe Ratio and Mean–Variance optimization criteria.

The purpose of this experiment is to assess whether allowing the macroeconomic representation and the information-state identification process to adapt continuously to newly available information provides additional benefits relative to the fixed feature representation employed in the baseline framework.

Table 4.3: Out-of-sample performance of the Rolling PCA framework (net of 10 bps transaction costs)

| Features   | Clusters | Strategy                   | Annual Return | Annual Vol. | Risk-Free | Sharpe | Alpha Ann. | Beta vs QQQ | Alpha t-stat | CAPM $R^2$ | VaR 95% | VaR 99% | ES 95%  | ES 99%  | Max DD  | Calmar |
|------------|----------|----------------------------|---------------|-------------|-----------|--------|------------|-------------|--------------|------------|---------|---------|---------|---------|---------|--------|
| RollingPCA | 4        | Mean_Variance              | 0.1464        | 0.1576      | 0.0383    | 0.6858 | 0.0679     | 0.3678      | 1.0300       | 0.2959     | -0.0138 | -0.0275 | -0.0229 | -0.0429 | -0.2293 | 0.6382 |
| RollingPCA | 3        | Max_Sharpe                 | 0.1418        | 0.1524      | 0.0383    | 0.6793 | 0.0529     | 0.4511      | 0.9681       | 0.4755     | -0.0145 | -0.0260 | -0.0216 | -0.0359 | -0.2670 | 0.5313 |
| RollingPCA | 3        | Mean_Variance              | 0.1171        | 0.1298      | 0.0383    | 0.6072 | 0.0316     | 0.4140      | 0.7427       | 0.5523     | -0.0126 | -0.0208 | -0.0185 | -0.0298 | -0.2207 | 0.5307 |
| Benchmark  | -        | Equal Weight               | 0.1114        | 0.1257      | 0.0383    | 0.5815 | 0.0224     | 0.4422      | 0.6377       | 0.6724     | -0.0125 | -0.0204 | -0.0174 | -0.0263 | -0.2085 | 0.5341 |
| RollingPCA | 4        | Max_Sharpe                 | 0.1353        | 0.1729      | 0.0383    | 0.5609 | 0.0485     | 0.4311      | 0.6968       | 0.3376     | -0.0156 | -0.0306 | -0.0254 | -0.0461 | -0.2725 | 0.4964 |
| RollingPCA | 5        | Mean_Variance              | 0.1290        | 0.1664      | 0.0383    | 0.5451 | 0.0506     | 0.3592      | 0.7108       | 0.2531     | -0.0140 | -0.0324 | -0.0248 | -0.0470 | -0.2547 | 0.5066 |
| RollingPCA | 2        | Mean_Variance              | 0.0991        | 0.1156      | 0.0383    | 0.5260 | 0.0238     | 0.3239      | 0.5561       | 0.4263     | -0.0118 | -0.0159 | -0.0229 | -0.2103 | 0.4714  |        |
| Benchmark  | -        | QQQ                        | 0.1535        | 0.2331      | 0.0383    | 0.4944 | 0.0000     | 1.0000      | 0.9277       | 1.0000     | -0.0237 | -0.0393 | -0.0331 | -0.0478 | -0.3483 | 0.4408 |
| Benchmark  | -        | SPY                        | 0.1254        | 0.1774      | 0.0383    | 0.4913 | 0.0032     | 0.7285      | 0.1289       | 0.9158     | -0.0170 | -0.0302 | -0.0254 | -0.0404 | -0.2450 | 0.5121 |
| RollingPCA | 2        | Max_Sharpe                 | 0.1038        | 0.1394      | 0.0383    | 0.4698 | 0.0177     | 0.4164      | 0.3628       | 0.4847     | -0.0139 | -0.0223 | -0.0193 | -0.0291 | -0.2519 | 0.4121 |
| Benchmark  | -        | BM_MeanVariance_NoClusters | 0.0871        | 0.1133      | 0.0383    | 0.4309 | 0.0103     | 0.3348      | 0.2580       | 0.4750     | -0.0114 | -0.0177 | -0.0154 | -0.0224 | -0.2114 | 0.4120 |
| RollingPCA | 5        | Max_Sharpe                 | 0.0893        | 0.1673      | 0.0383    | 0.3049 | 0.0068     | 0.3843      | 0.0987       | 0.2865     | -0.0153 | -0.0312 | -0.0251 | -0.0435 | -0.2710 | 0.3296 |
| Benchmark  | -        | BM_Max_Sharpe_NoClusters   | 0.0545        | 0.1333      | 0.0383    | 0.1215 | -0.0348    | 0.4480      | -0.8884      | 0.6145     | -0.0140 | -0.0204 | -0.0182 | -0.0250 | -0.2990 | 0.1822 |

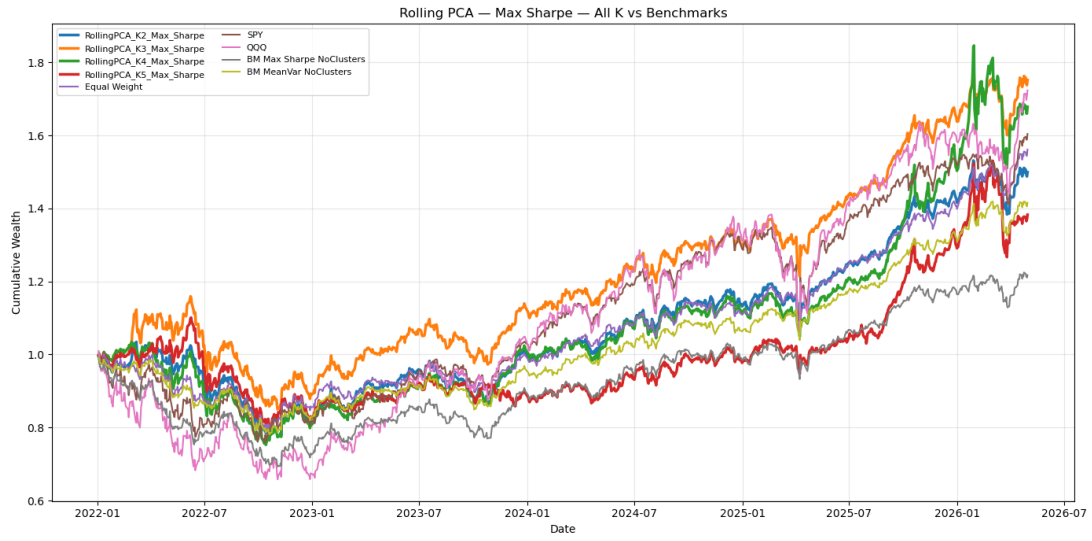


Figure 4.14: Out-of-sample cumulative wealth of the best Rolling PCA specification under the Maximum Sharpe criterion ( $K = 3$ ), compared with the benchmark portfolios.

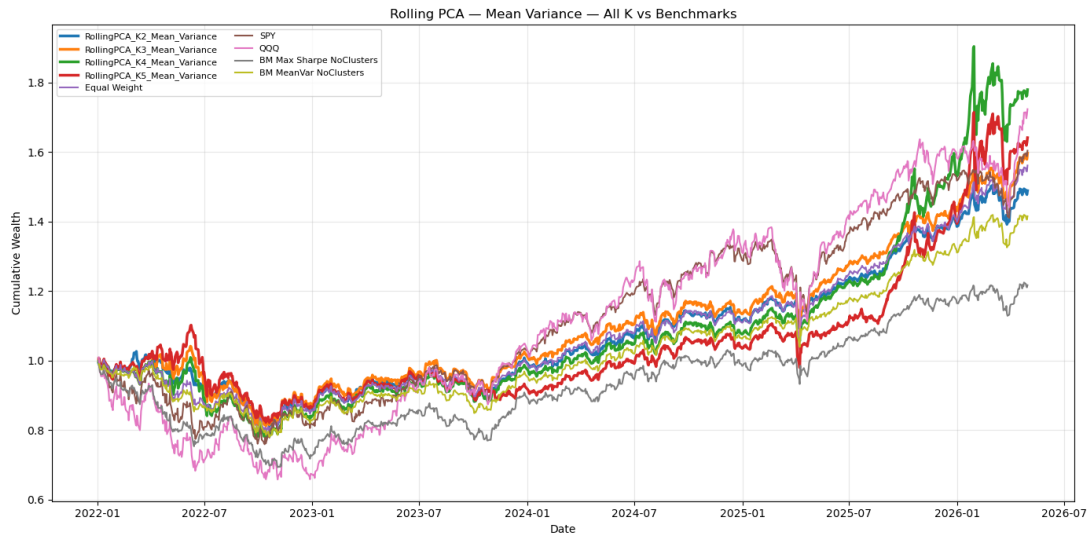


Figure 4.15: Out-of-sample cumulative wealth of the best Rolling PCA specification under the Mean–Variance criterion ( $K = 4$ ), compared with the benchmark portfolios.

### 4.3.1 Discussion of the Rolling PCA Results

The Rolling PCA framework extends the baseline Expanding specification by recursively updating not only the portfolio optimization stage but also the feature extraction process. At each rebalancing date, the principal components are re-estimated using all macroeconomic observations available up to that point, allowing the latent representation of the macroeconomic environment to evolve over time.

The empirical results indicate that this additional degree of flexibility increases the annual returns achieved by several model specifications. The best-performing configuration is obtained under the Mean–Variance optimization with four information states, which achieves an annual return of 14.64%, an annualized volatility of 15.76% and a Sharpe ratio of 0.686. The corresponding Maximum Sharpe specification with three information states also performs well, recording an annual return of 14.18% together with a Sharpe ratio of 0.679.

As emphasized throughout this work, the objective of the proposed methodology is not to maximize absolute returns or to systematically outperform the market. Indeed, both the SPY and, in particular, the QQQ benchmark achieve higher annual returns than most of the Rolling PCA specifications. However, these higher returns are associated with substantially greater portfolio risk, leading to lower Sharpe ratios.

This pattern is also evident across the Rolling PCA strategies. Only two specifications record a Sharpe ratio below those of the market benchmarks, whereas all the remaining configurations achieve superior risk-adjusted performance. These results suggest that the proposed framework is generally capable of constructing portfolios with a more efficient risk–return trade-off than conventional passive equity investments.

Compared with the baseline framework, the Rolling PCA methodology primarily improves annual returns rather than risk-adjusted performance. Although several specifications outperform their baseline counterparts in terms of return, the highest Sharpe ratio remains that of the baseline PCA specification with three information states under the Expanding Maximum Sharpe framework. Therefore, recursively updating the principal-component representation appears to increase the responsiveness of the model to evolving macroeconomic conditions, but the additional flexibility is accompanied by higher portfolio volatility, preventing a systematic improvement in Sharpe ratios.

Overall, the comparison highlights an important trade-off. While the Rolling PCA framework enhances the model’s ability to adapt to changes in the macroeconomic environment and generally increases portfolio returns, the baseline framework continues to provide the highest level of risk-adjusted performance. This result suggests that increasing model flexibility does not necessarily translate into superior portfolio efficiency, although the Rolling PCA approach remains competitive relative to traditional market benchmarks.

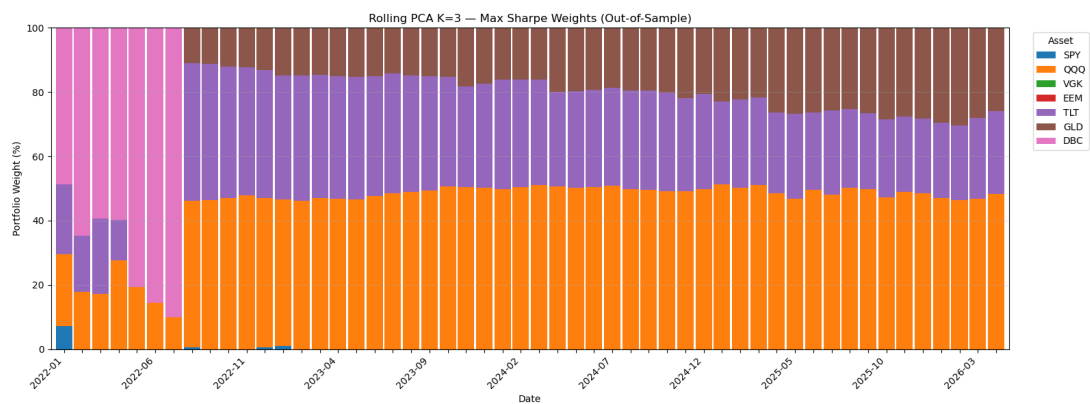


Figure 4.16: Out-of-sample portfolio weights obtained under the Rolling PCA framework using the Maximum Sharpe optimization criterion with three information states ( $K = 3$ ).

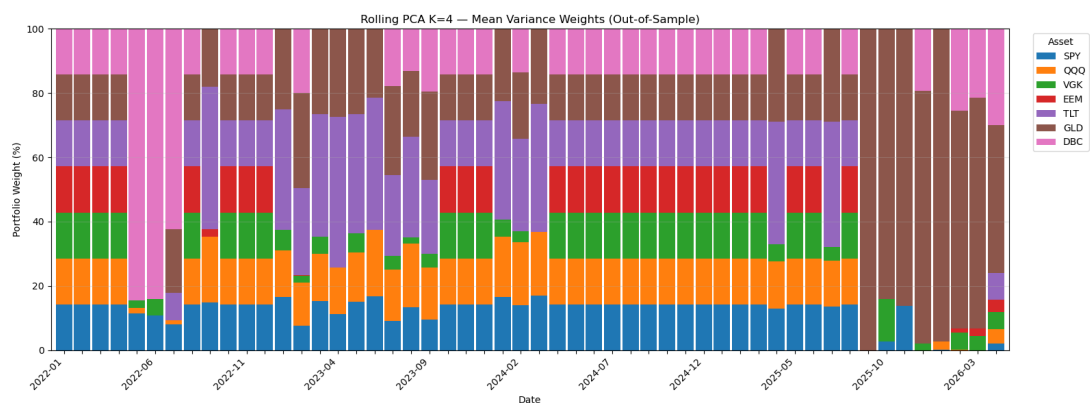


Figure 4.17: Out-of-sample portfolio weights obtained under the Rolling PCA framework using the Mean–Variance optimization criterion with four information states ( $K = 4$ ).

Figures 4.16 and 4.17 report the out-of-sample portfolio allocations associated with the two best-performing Rolling PCA specifications according to the Sharpe ratio. The first figure refers to the Maximum Sharpe portfolio with three information states, while the second reports the Mean–Variance portfolio with four information states.

Both strategies exhibit substantial time variation in their portfolio composition, reflecting the recursive updating of both the principal-component representation and the optimization inputs. Compared with the baseline framework, the portfolio weights adjust more frequently over time, indicating that the Rolling PCA approach produces a more responsive asset allocation as new macroeconomic information becomes available.

Despite these differences, both portfolios remain well diversified across the available asset classes throughout most of the evaluation period. The changing asset allocations illustrate how the dynamic feature extraction process translates into different investment decisions as the macroeconomic environment evolves.

### 4.3.2 CAPM Analysis

The CAPM statistics indicate that most Rolling PCA specifications generate positive annualized alpha estimates, suggesting that the dynamic re-estimation of the macroeconomic feature representation is associated with returns exceeding those implied by the strategies' systematic exposure to the QQQ benchmark. In particular, the two best-performing specifications report annualized alpha values of 6.79% and 5.29%, respectively, while maintaining beta coefficients well below one, indicating a substantially lower exposure to market risk than the benchmark.

Despite these favourable alpha estimates, the corresponding alpha t-statistics remain below conventional significance thresholds. Consequently, the null hypothesis of zero abnormal performance cannot be rejected, implying that the observed positive alpha may reflect sampling variability rather than persistent outperformance.

Compared with the baseline framework, the Rolling PCA specifications generally preserve positive alpha estimates while operating with relatively low beta values, suggesting that the recursive updating of the feature representation does not increase systematic market exposure. Overall, the CAPM results confirm that the Rolling PCA framework remains capable of delivering competitive risk-adjusted performance, although the evidence is insufficient to conclude that it systematically generates statistically significant abnormal returns.

The estimated coefficients of determination are generally lower than those observed for the baseline framework, ranging from approximately 0.28 to 0.58 across the different Rolling PCA specifications. This suggests that a smaller proportion of the variability in portfolio excess returns is explained by the CAPM, indicating that the dynamically updated strategies are less closely tied to the QQQ benchmark. Consequently, the recur-

sive updating of the macroeconomic feature representation appears to produce portfolio returns that are relatively more independent of broad market movements, while still maintaining competitive risk-adjusted performance.

## 4.4 Annual Adaptive Re-estimation Framework

The final extension of the proposed methodology introduces an Adaptive Annual Re-estimation Framework. Unlike the baseline specification, where the information states remain fixed throughout the entire out-of-sample period, and the Rolling PCA framework, where only the feature representation is recursively updated, this extension allows the entire portfolio construction procedure to adapt over time through an annual model selection process.

At the end of each calendar year, the complete set of candidate models is re-estimated using exclusively the information available up to December of that year. This includes all alternative clustering configurations considered throughout the analysis, together with both portfolio optimization criteria. For each candidate specification, the corresponding out-of-sample performance over the previous year is evaluated using the Sharpe ratio as the model selection criterion.

The specification that achieves the highest Sharpe ratio over the evaluation period is then selected and implemented throughout the following calendar year. Consequently, the number of information states, the estimated cluster centroids, the Linear Discriminant Analysis classifier and the associated portfolio weights are all updated annually using only the information available at the time the investment decision is made.

By updating the entire framework only once per year, the proposed methodology seeks to strike a balance between adaptability and stability. On the one hand, the model is allowed to react to gradual structural changes in the macroeconomic environment by periodically revising the clustering structure and the portfolio allocation. On the other hand, the annual re-estimation frequency avoids the excessive parameter instability and portfolio turnover that may arise when the model is re-estimated at every monthly rebalancing date.

The objective of this extension is therefore not to identify a universally superior specification, but rather to mimic the behaviour of a practitioner who periodically reviews the historical performance of competing models and selects the specification that has provided the most favourable risk-adjusted performance based solely on the information available at that point in time. The empirical analysis presented in the following sections evaluates whether this adaptive model selection procedure is capable of improving portfolio performance relative to both the baseline and the Rolling PCA frameworks.

Table 4.4: Out-of-sample performance of the Adaptive strategy, the five best fixed-clustering models and the market benchmarks (net of 10 bps transaction costs).

| Features  | Clusters | Strategy                       | Annual Return | Annual Vol. | Risk-Free | Sharpe | Alpha Ann. | Beta vs QQQ | Alpha t-stat | CAPM $R^2$ | VaR 95% | VaR 99% | ES 95%  | ES 99%  | Max DD  | Calmar |
|-----------|----------|--------------------------------|---------------|-------------|-----------|--------|------------|-------------|--------------|------------|---------|---------|---------|---------|---------|--------|
| PCA       | 3        | Expanding_Max_Sharpe           | 0.1382        | 0.1403      | 0.0383    | 0.7122 | 0.0461     | 0.4760      | 1.0878       | 0.6251     | -0.0147 | -0.0220 | -0.0195 | -0.0288 | -0.2304 | 0.5999 |
| PCA       | 4        | Expanding_Max_Sharpe           | 0.1271        | 0.1406      | 0.0383    | 0.6316 | 0.0323     | 0.4942      | 0.8188       | 0.6714     | -0.0140 | -0.0224 | -0.0196 | -0.0289 | -0.2326 | 0.5464 |
| RAW       | 3        | Expanding_Mean_Variance        | 0.1166        | 0.1250      | 0.0383    | 0.6266 | 0.0288     | 0.4331      | 0.7986       | 0.6516     | -0.0123 | -0.0204 | -0.0174 | -0.0263 | -0.2058 | 0.5667 |
| RAW       | 3        | Static_Mean_Variance           | 0.1153        | 0.1251      | 0.0383    | 0.6153 | 0.0272     | 0.4351      | 0.7590       | 0.6572     | -0.0125 | -0.0204 | -0.0174 | -0.0263 | -0.2058 | 0.5600 |
| PCA       | 3        | Expanding_Mean_Variance        | 0.1114        | 0.1257      | 0.0383    | 0.5815 | 0.0224     | 0.4421      | 0.6376       | 0.6723     | -0.0125 | -0.0204 | -0.0174 | -0.0263 | -0.2085 | 0.5341 |
| Benchmark | –        | Equal Weight                   | 0.1114        | 0.1257      | 0.0383    | 0.5815 | 0.0224     | 0.4422      | 0.6377       | 0.6724     | -0.0125 | -0.0204 | -0.0174 | -0.0263 | -0.2085 | 0.5341 |
| Adaptive  | –        | Yearly_Rolling_KMeans_Selected | 0.1051        | 0.1234      | 0.0383    | 0.5410 | 0.0207     | 0.4017      | 0.5278       | 0.5754     | -0.0120 | -0.0183 | -0.0165 | -0.0256 | -0.2066 | 0.5087 |
| Benchmark | –        | SPY                            | 0.1254        | 0.1774      | 0.0383    | 0.4913 | 0.0032     | 0.7285      | 0.1289       | 0.9158     | -0.0170 | -0.0302 | -0.0254 | -0.0404 | -0.2450 | 0.5121 |
| Benchmark | –        | QQQ                            | 0.1535        | 0.2331      | 0.0383    | 0.4944 | 0.0000     | 1.0000      | 0.9277       | 1.0000     | -0.0237 | -0.0393 | -0.0331 | -0.0478 | -0.3483 | 0.4408 |

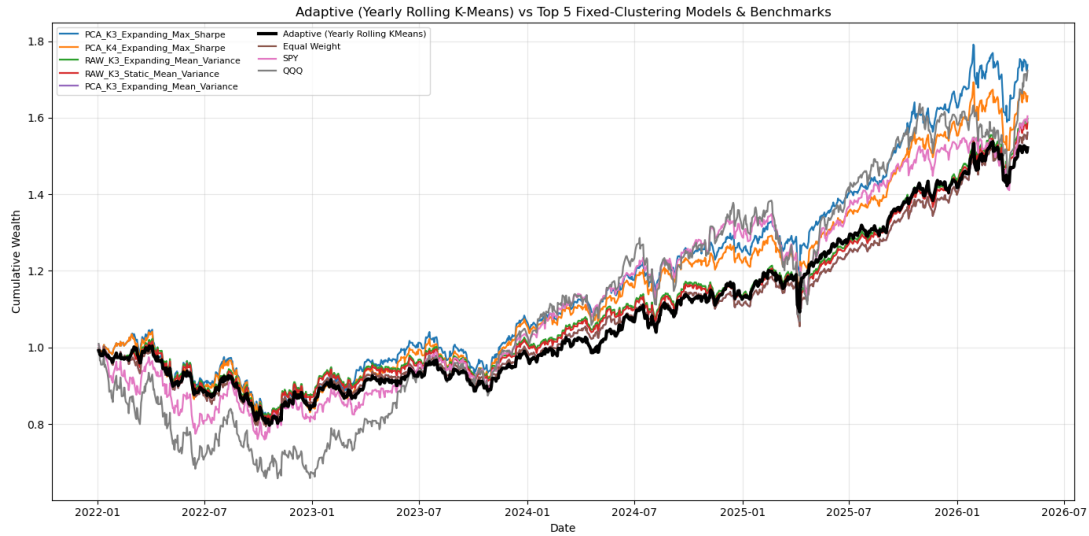


Figure 4.18: Out-of-sample cumulative wealth of the Adaptive (Yearly Rolling K-Means) strategy compared with the best fixed-clustering strategies and the benchmark portfolios. Out-of-sample cumulative wealth of the Adaptive (Yearly Rolling K-Means) strategy compared with the best fixed-clustering strategies and the benchmark portfolios.

#### 4.4.1 Discussion of the Adaptive Framework Results

The Adaptive Annual Re-estimation strategy achieves an annual return of 10.51% with an annualized volatility of 12.34%, resulting in a Sharpe ratio of 0.541. Although its absolute return is lower than those of both SPY and QQQ, the strategy operates with substantially lower volatility and achieves a higher Sharpe ratio than both market benchmarks, whose corresponding values are 0.491 and 0.494, respectively. The strategy also exhibits a maximum drawdown of 20.66%, compared with 24.50% for SPY and 34.83% for QQQ.

The Adaptive specification does not dominate the best baseline configurations in terms of risk-adjusted performance, ranking 13th among the 38 strategies considered according to the Sharpe ratio. However, its main contribution should not be evaluated exclusively through this ex-post ranking. Unlike the fixed model specifications, the Adaptive framework does not require the investor to select ex ante a specific feature representation, number of information states, or portfolio optimization rule and maintain that choice throughout the entire investment horizon.

Instead, the framework periodically reassesses the available candidate models using only information available at the time and implements in the following year the specification that achieved the strongest risk-adjusted performance during the preceding evaluation period. This characteristic represents a strategic advantage of the methodology: the investment process is not structurally dependent on a single model configuration, but is able to modify its allocation rule as the empirical performance of the competing specifications evolves.

From this perspective, the Adaptive framework can be interpreted as a model-selection layer placed above the individual portfolio strategies. Its value therefore lies not only in the observed out-of-sample performance, but also in providing a systematic and fully rule-based mechanism for responding to changes in the relative effectiveness of alternative portfolio construction models over time.

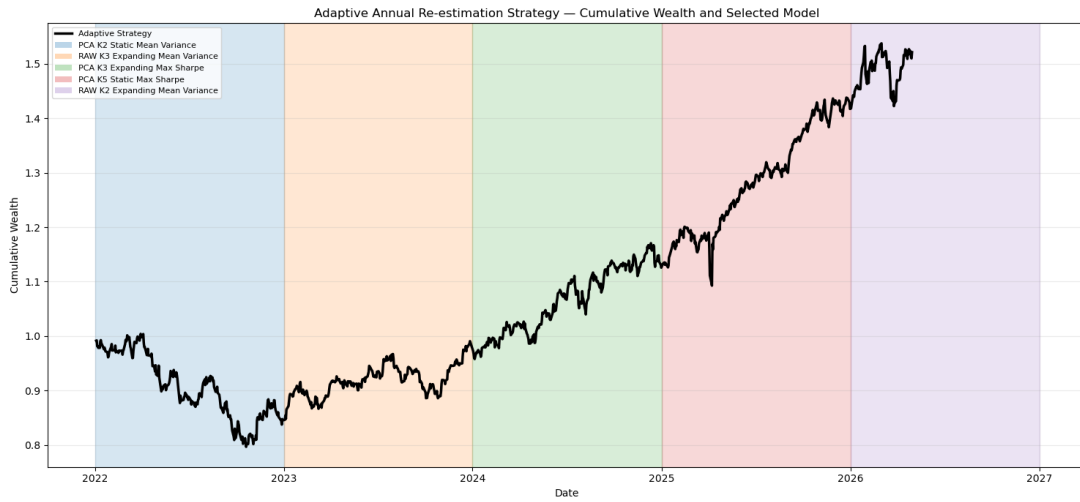


Figure 4.19: Out-of-sample cumulative wealth of the Adaptive Annual Re-estimation strategy. The background colors indicate the portfolio specification selected at the beginning of each calendar year based exclusively on the information available up to the end of the previous year.

Figure 4.19 illustrates the behavior of the adaptive allocation strategy throughout the out-of-sample period. Each colored region identifies the portfolio specification selected for the corresponding calendar year according to the historical performance observed up to the end of the previous year. Consequently, the strategy continuously adapts its investment rule without relying on any future information, thereby preserving the out-of-sample nature of the experiment. This figure highlights the key feature of the proposed framework: rather than committing to a single model for the entire investment horizon, the strategy dynamically switches among alternative portfolio specifications as market conditions evolve.

### 4.4.2 CAPM Analysis

The Adaptive strategy reports an annualized alpha of 2.07% together with a beta of 0.402, indicating that the portfolio achieves a modest positive abnormal return while maintaining a substantially lower exposure to market risk than the QQQ benchmark. The corresponding alpha t-statistic is equal to 0.53, well below conventional significance thresholds, implying that the estimated alpha is not statistically different from zero.

The estimated coefficient of determination ( $R^2 = 0.575$ ) indicates that approximately 58% of the variability of the portfolio excess returns is explained by movements in the QQQ benchmark, while the remaining variation reflects idiosyncratic components not captured by the CAPM. Compared with the market benchmarks, this relatively moderate  $R^2$  is consistent with the objective of constructing a diversified portfolio whose performance is not entirely driven by broad equity market movements.

Overall, the CAPM analysis suggests that the Adaptive framework preserves a relatively low level of systematic market exposure while generating a positive estimated alpha. However, as in the baseline and Rolling PCA frameworks, the empirical evidence does not support the existence of statistically significant abnormal returns after controlling for market risk.

## 4.5 Further Developments

The empirical results presented in the previous sections suggest several directions through which the proposed framework could be further developed. In particular, two extensions appear especially relevant: increasing the timeliness and richness of the information set used by the model, and introducing an explicit derivative-based layer for downside-risk management.

A first direction concerns the use of more institutional and real-time data infrastructures. The current empirical framework relies mainly on historical macroeconomic series observed at a monthly frequency. A natural development would be to connect the model directly to institutional data providers, allowing new observations to enter the information set as soon as they become available. This would make the investment process more responsive to changes in the economic environment and reduce the gap between the historical backtesting framework and a real-time implementation.

A further development could incorporate unstructured information, particularly financial and geopolitical news. Natural Language Processing techniques could be used to transform real-time news flows into quantitative features describing the type, intensity and economic relevance of newly occurring events. Clustering techniques could then be applied to these news-based representations in order to identify recurring patterns of information and associate them with the macroeconomic variables or asset classes most

likely to be affected. Such an extension would allow the framework to react not only to periodically released macroeconomic indicators, but also to unexpected events that may rapidly alter market expectations and asset-price dynamics.

This approach could be integrated with the existing information-state framework by augmenting the monthly macroeconomic dataset with higher-frequency signals extracted from news and other real-time sources. The resulting model would therefore combine structured macroeconomic information with event-driven indicators, potentially improving the speed with which changes in the economic information environment are reflected in portfolio allocation decisions.

A second development concerns explicit downside-risk management through derivatives. In particular, the information states identified by the model could be used to determine when an option-based hedging overlay should be activated. One possible implementation would be a dynamic collar strategy, constructed by combining the underlying portfolio with the purchase of a protective put option and the simultaneous sale of a call option.

$$V_{\text{Collar}} = S + P(K_P, T) - C(K_C, T), \quad (4.1)$$

where  $K_P$  and  $K_C$  denote the strike prices of the OTM put and call options, respectively.

To illustrate the economic intuition of the strategy, consider a simplified example in which the QQQ ETF, used throughout the empirical analysis as the reference market benchmark, has an initial price equal to

$$S_0 = 100.$$

Suppose that the investor expects a potentially large movement in the underlying price and therefore implements a collar by purchasing an out-of-the-money put option with strike price  $K_P = 70$  and simultaneously selling an out-of-the-money call option with strike price  $K_C = 130$ . For simplicity, the put premium is assumed to be exactly offset by the premium received from selling the call, resulting in a zero-cost collar.

Under this assumption, the profit and loss of the strategy at maturity can be written as

$$\Pi_{\text{Collar}}(S_T) = \min[\max(S_T, K_P), K_C] - S_0. \quad (4.2)$$

The collar therefore limits both downside losses and upside gains. If the QQQ price falls to  $S_T = 50$ , the protective put becomes effective and guarantees a terminal portfolio value of 70. The resulting profit and loss is therefore

$$\Pi_{\text{Collar}}(50) = 70 - 100 = -30,$$

compared with a loss of 50 that would have been experienced by holding the underlying asset without protection.

Conversely, if the QQQ price increases to  $S_T = 150$ , the short call limits the terminal value of the position to 130. The corresponding profit is therefore

$$\Pi_{\text{Collar}}(150) = 130 - 100 = +30,$$

instead of the +50 gain that would have been obtained from an unhedged position. The collar therefore reduces exposure to extreme downside movements at the cost of giving up part of the potential upside.

If the underlying price remains between the two strike prices at maturity, neither the put nor the call is exercised and the investor simply retains the underlying position. In this case, the only potential cost of the hedge would be the net option premium paid to establish the collar. Under the simplifying zero-cost assumption adopted here, this cost is equal to zero because the put premium is assumed to be exactly offset by the premium received from selling the call.

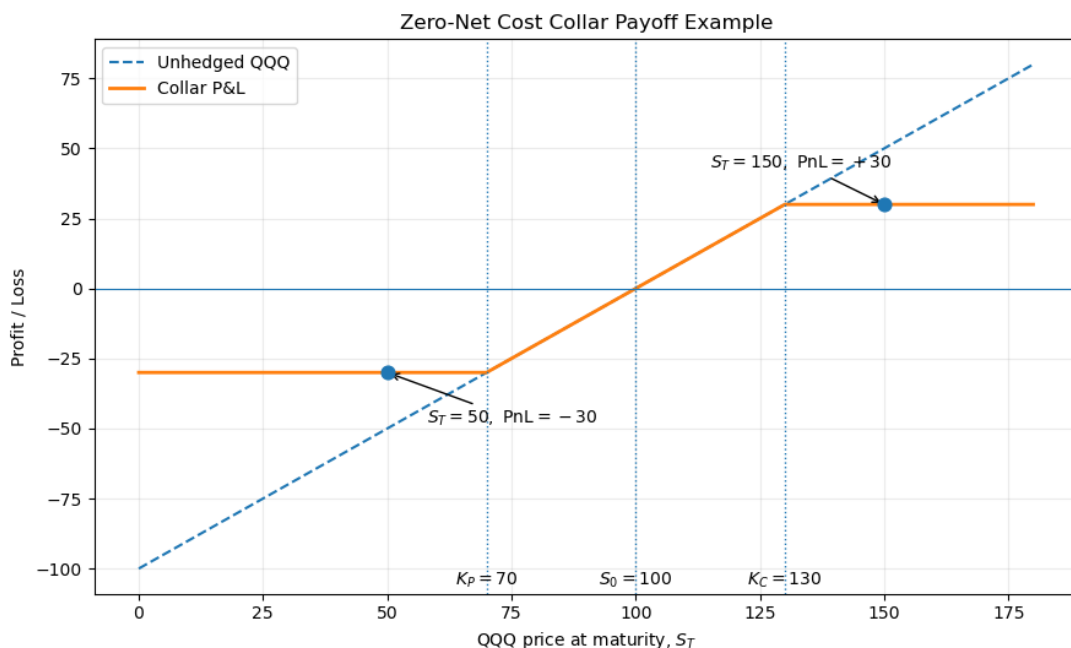


Figure 4.20: Illustrative profit-and-loss profile of a zero-cost collar on the QQQ ETF, assuming  $S_0 = 100$ ,  $K_P = 70$ , and  $K_C = 130$ . The dashed line represents the unhedged position, while the collar limits both the maximum loss and the maximum gain to 30 monetary units relative to the initial investment.

The intensity of the hedge could depend on the level of estimated portfolio risk or on the prevailing information state. For example, more defensive collar structures could be employed when the model identifies states associated with elevated downside risk, while less restrictive or no option protection could be used under more favourable conditions.

This would preserve the existing asset-allocation framework while introducing an additional layer specifically designed to limit extreme losses.

Such an extension would require reliable historical option data, including strike prices, maturities, implied volatilities and transaction costs. It would therefore represent a substantially more data-intensive empirical exercise, but it could provide a natural continuation of the present research by investigating whether the information states can be used not only to determine asset allocation, but also to dynamically control portfolio downside exposure.

# Chapter 5

## Conclusions

### 5.1 Conclusion remarks

This thesis developed a machine-learning-based framework for dynamic portfolio allocation in which macroeconomic information is first organized into latent information states and subsequently translated into portfolio decisions. The methodological contribution lies primarily in the combination of unsupervised and supervised learning techniques used to characterize the information environment, while portfolio optimization provides the financial application through which the usefulness of the resulting states is evaluated.

The empirical analysis considered alternative feature representations, different numbers of information states, two portfolio optimization criteria and several updating schemes. Overall, the results indicate that conditioning portfolio construction on the identified information states can generate allocations with competitive out-of-sample risk-adjusted performance. In particular, several specifications achieve Sharpe ratios above those of the SPY and QQQ benchmarks while operating with substantially lower volatility and drawdowns. These findings should not be interpreted as evidence of systematic market outperformance in terms of absolute returns. Rather, they suggest that the proposed framework can provide a more favourable balance between return and risk under certain model configurations.

Among the baseline specifications, the strongest out-of-sample result is obtained by the PCA representation with three information states under the Expanding Maximum Sharpe framework. This configuration achieves the highest Sharpe ratio among the strategies considered in the baseline analysis. More generally, the Expanding specifications tend to perform better than their Static counterparts, indicating that updating the state-specific expected returns and covariance matrices as new financial observations become available improves the adaptability of the portfolio construction process while preserving a fixed representation of the information states.

The Rolling PCA framework provides a complementary result. By recursively re-estimating the standardization parameters, the principal-component representation and the K-Means clustering structure, the model becomes more responsive to changes in the macroeconomic information environment. Several Rolling PCA specifications achieve higher annual returns than their baseline counterparts. However, this additional flexibility is accompanied by higher volatility, and the highest Sharpe ratio remains associated with

the baseline Expanding framework. The comparison therefore highlights an important trade-off between adaptability and stability: increasing the frequency with which the information representation is updated can improve responsiveness, but does not necessarily lead to superior risk-adjusted performance.

The Annual Adaptive Re-estimation framework addresses the same trade-off from a different perspective. Rather than committing to a single model specification for the entire out-of-sample period, the framework periodically evaluates the competing strategies using only historically available information and selects the specification with the strongest recent risk-adjusted performance for implementation in the following year. Although the Adaptive strategy does not dominate the best fixed specifications *ex post*, its main contribution is strategic rather than purely performance-based. It provides a systematic rule for changing the model configuration over time and reduces the dependence of the investment process on a single *ex-ante* modelling choice.

For this reason, the Adaptive framework can be regarded as the most robust specification examined in this thesis, since it does not rely on maintaining a single model configuration throughout the entire out-of-sample period. Instead, it periodically reassesses the available alternatives and allows the investment rule to evolve as their relative risk-adjusted performance changes over time.

The CAPM analysis provides additional evidence regarding the nature of the observed portfolio returns. Most of the better-performing specifications exhibit beta coefficients substantially below one, indicating lower systematic exposure to the QQQ benchmark. Several strategies also generate positive annualized alpha estimates. However, the corresponding alpha t-statistics remain below conventional significance thresholds, and therefore the empirical evidence does not support the conclusion that the proposed framework systematically generates statistically significant abnormal returns. The estimated coefficients of determination further indicate that only part of the variation in portfolio excess returns is explained by exposure to the QQQ benchmark, particularly for the more dynamically updated specifications.

The results should nevertheless be interpreted in light of several simplifying assumptions. First, the empirical analysis assumes that monthly macroeconomic information is available in a timely and homogeneous manner, while in practice many macroeconomic series are released with delays and subsequently revised. The use of revised historical data therefore does not fully reproduce the real-time information set available to an investor. Second, transaction costs are represented through a constant proportional cost of 10 basis points, while actual trading costs may vary across assets and market conditions. Market impact, liquidity constraints, taxation, slippage and execution delays are also excluded from the analysis. Finally, the empirical evidence is necessarily conditional on the selected sample period, investment universe and set of macroeconomic variables.

Overall, the empirical evidence suggests that machine-learning techniques can provide

a useful structure for transforming a heterogeneous macroeconomic information set into systematic portfolio allocation decisions. The main result of the thesis is therefore not the identification of a single universally superior investment strategy, but the demonstration that information-state-based portfolio construction can offer a flexible and interpretable framework for adapting asset allocation to changes in the economic information environment while maintaining an explicit focus on out-of-sample risk-adjusted performance.

# Bibliography

- [1] Saeed Aghabozorgi, Ali Seyed Shirkhorshidi, and Teh Ying Wah. Time-series clustering – a decade review. *Information Systems*, 53:16–38, 2015.
- [2] Andrew Ang and Allan Timmermann. Regime changes and financial markets. *Annual Review of Financial Economics*, 4:313–337, 2012.
- [3] Matthew F. Dixon, Igor Halperin, and Paul Bilokon. *Machine Learning in Finance: From Theory to Practice*. Springer, 2020.
- [4] Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep Learning*. MIT Press, 2016.
- [5] Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5):2223–2273, 2020.
- [6] Massimo Guidolin and Allan Timmermann. Asset allocation under multivariate regime switching. *Journal of Economic Dynamics and Control*, 31(11):3503–3544, 2007.
- [7] James D. Hamilton. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2):357–384, 1989.
- [8] Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer, 2 edition, 2009.
- [9] Harold Hotelling. Analysis of a complex of statistical variables into principal components. *Journal of Educational Psychology*, 24(6):417–441, 1933.
- [10] Anil K. Jain. Data clustering: 50 years beyond k-means. *Pattern Recognition Letters*, 31(8):651–666, 2010.
- [11] Ian T. Jolliffe. *Principal Component Analysis*. Springer, 2 edition, 2002.
- [12] Bryan T. Kelly, Seth Pruitt, and Yinan Su. Characteristics are covariances: A unified model of risk and return. *Journal of Financial Economics*, 134(3):501–524, 2019.
- [13] Stuart Lloyd. Least squares quantization in pcm. *IEEE Transactions on Information Theory*, 28(2):129–137, 1982.
- [14] Marcos López de Prado. *Advances in Financial Machine Learning*. Wiley, 2018.

- [15] James MacQueen. Some methods for classification and analysis of multivariate observations. In *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*, volume 1, pages 281–297, 1967.
- [16] Peter Nystrup, Bo William Hansen, Henrik Olejasz Larsen, Henrik Madsen, and Erik Lindström. Dynamic allocation or diversification: A regime-based approach to multiple assets. *The Journal of Portfolio Management*, 44(2):62–73, 2018.
- [17] Karl Pearson. On lines and planes of closest fit to systems of points in space. *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, 2(11):559–572, 1901.